

Economic Consequences of Sequence-Dependent Betting and Its Psychological Underpinnings

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Abstract

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1 Introduction

The 1913 legend has it that a roulette wheel in Monte Carlo produced a streak of 26 blacks, prompting gamblers to lose millions of francs betting on reversal — convinced that red was overdue. Yet what the legend leaves unclear is how exactly those losses were generated. Did everyone bet on reversal, consistent with what Lepley (1963) later called the *gambler’s fallacy*? Or were there others who read the same streak as evidence of a biased wheel and bet on continuation — consistent with the *hot-hand fallacy* (Gilovich et al., 1985)? Did the streak increase the perceived chance of winning so that even the most risk-averse observers decided to place a bet? And were people staking more than usual, reflecting their confidence in whichever direction they chose? The historical record is silent on all of this. Decades of research have, however, since shed considerable light on both fallacies.

The early account traced them to a common root: the misconception that chance is a self-correcting process (Kahneman & Tversky, 1972; Tversky & Kahneman, 1974) makes streaks look suspiciously non-random. It is then a matter of how one resolves the suspicion. The gambler’s fallacy (GF) resolves it by expecting a correction: the streak is seen as a temporary deviation that randomness must soon undo, making the opposite outcome more likely (Tversky & Kahneman, 1971). The hot-hand fallacy (HF) resolves it the other way: the streak is too long to be mere chance and must reflect some underlying force, making continuation expected (Gilovich et al., 1985).¹ The theoretical frameworks of Rabin

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¹ The analysis of Gilovich et al. (1985) rested on treating shooting as an independent Bernoulli process — an assumption later shown to contain a selection bias that, once corrected, reverses the conclusions of the canonical study, with bias-corrected effect sizes consistently moderate to large across controlled shooting datasets (Miller & Sanjurjo, 2018, 2024). Evidence across other sports remains mixed: significant streakiness has been documented in baseball (Green & Zwiebel, 2018), tennis (Klaassen & Magnus, 2001), and volleyball (Burns & Corpus, 2004; Raab et al., 2012), while analyses of NHL goalies (Ding et al., 2025) and professional darts (Ötting et al., 2020) found no conclusive evidence, suggesting performance patterns are task-specific (Morgulev & Avugos, 2023; Raab & MacMahon, 2015). The present study concerns beliefs about objectively random sequences and is orthogonal to this debate.

(2002) and Rabin and Vayanos (2010) provide the most rigorous account of when the same underlying misconception resolves as GF or HF. In the urn model of Rabin (2002), the observer falsely believes signals are drawn without replacement from a finite urn that must eventually balance out — generating GF for short streaks, as the urn is perceived as depleting. When a streak persists long enough to surprise the observer, however, they over-infer that the underlying urn rate has shifted toward the more frequent outcome, flipping expectations from reversal to continuation. Rabin and Vayanos (2010) extend this logic to a richer signal-state framework, showing that GF dominates short streaks while HF emerges after long ones as the agent over-infers a persistent shift in the underlying state. Importantly, as Gilovich et al. (1985) already noted, this misperception operates even when sequences are displayed statically rather than experienced sequentially — ruling out memory salience as the sole mechanism and suggesting the error originates at the inference stage itself. A formally distinct but empirically convergent account was recently offered by Dorst (2026), who demonstrates that the same nonlinear pattern emerges from rational Bayesian agents who begin with causal uncertainty about whether a process is independent, “switchy” (tending toward alternation), or “sticky” (tending toward streaks). Because switchy distributions are statistically closer to the independent distribution than sticky ones are, unbiased data asymmetrically rules out the latter faster; agents therefore remain on average more confident of a switchy process, producing a persistent predisposition toward gambler’s reasoning without any model misspecification. Unlike Rabin’s framework, which requires agents to hold an incorrect causal model, Dorst’s account attributes the bias to bounded-optimal inference under limited data and memory. The empirical record speaks to both accounts.

In the laboratory, Asparouhova et al. (2009) elicit incentive-compatible probability assessments across 100 randomly generated eight-outcome binary sequences, finding the predicted non-monotonic relationship: subjects expect reversals after short streaks and continuation after long ones, with the Rabin model fitting significantly better than both a regime-shifting alternative and a Bayesian benchmark. Carlson and Shu (2007) further identify the third repeat outcome as the empirical pivot point at which perceived streakiness plateaus and observers begin to infer a genuine streak rather than noise. Importantly, Oskarsson et al. (2009) document substantial individual heterogeneity in these judgments: in a representative sample, roughly a third of participants exhibit GF, a third respond normatively, and a fifth exhibit HF — suggesting that aggregate measures mask meaningfully distinct response patterns across individuals. In the field, Croson and Sundali (2005) and Sundali and Croson (2006) use videotaped roulette play to show that both fallacies coexist simultaneously — players exhibiting GF when betting on outcomes and HF regarding their own luck — with Sundali and Croson (2006) documenting a significant within-person correlation between the two biases, consistent with a common underlying cause. Suetens et al. (2016) identify a pattern in longitudinal Danish lottery data where players avoid recently drawn numbers after short runs but shift toward betting on them as streaks lengthen, though Dillon and Lybbert (2024) challenge this finding using player-level administrative data from both Haiti and Denmark, showing the pattern disappears after correcting a linearity assumption — with the average player consistently avoiding recently winning numbers (GF) and only a small subset betting on them (HF). In financial markets, Pelster (2020) finds the same non-monotonic pattern in retail investors’ trading propensity, directly consistent with Rabin and Vayanos (2010).

Beyond streak length, the perceived provenance of the sequence significantly modulates the resolution of sequential biases. Ayton and Fischer (2004) identifies causal attribution as the primary determinant: while sequences attributed to inanimate chance mechanisms induce expectations of reversal (GF), those linked to human performance or skill elicit expectations of continuation (HF). This is experimentally corroborated by Burns and Corpus (2004), who demonstrate that the degree of a scenario’s perceived nonrandomness determines which fallacy dominates: truly random mechanisms

such as a roulette wheel elicit reversal expectations, while nonrandom scenarios such as competitive interactions elicit streak continuation expectations. Caruso et al. (2010) further extends this mechanism, establishing that the mere perception of intentionality is sufficient to induce HF; observers predict streak continuation when they anthropomorphize a generator, even when the underlying data-generating process is held constant. The instability of these attributions is highlighted by Huber et al. (2010), who find that while participants predicting coin tosses for themselves exhibit GF, simply labeling a randomized generator as an “expert” is enough to trigger HF, as participants select advisors based on recent streaks of success. Recent evidence suggests that these generator-specific effects are mediated by uncertainty regarding the underlying parameters. Rao and Hastie (2023) demonstrate that providing participants with an explicit distribution of possible base rates virtually eliminates generator-driven differences — suggesting the bias operates through uncertainty about the underlying process rather than the generator’s identity per se, and consistent with Dorst (2026), who formalises this mechanism: on his account, GF reasoning persists precisely because agents remain uncertain whether the process is stationary, and dissipates as that uncertainty resolves. Finally, the choice environment’s structure serves as a critical moderator independent of the generator’s identity. Kong et al. (2020) provide evidence that the density of reward realizations — the number of prizes drawn per round — dictates the dominant fallacy: holding the mechanical randomizer constant, sparse reward realizations induce GF, whereas a high frequency of winning outcomes leads observers to revise their priors toward continuation, flipping aggregate behavior toward HF.

These findings collectively establish that the resolution of sequential biases depends on both streak properties and the features of the environment. A separate but equally important question is whether these biases carry meaningful economic consequences. Ball (2012) offers an early warning: risk preferences during runs of gains and losses vary not only with run valence and length but also with the decision maker’s current financial state — a wealth confound that field studies cannot control for and that prior lab studies had largely overlooked. More direct evidence of costs comes from casino settings: Muto et al. (2025) find that baccarat gamblers predominantly follow trends, with bet amounts increasing slightly but reliably with streak length; crucially, trend-following on tie bets — which carry a 9.5% house advantage — generates nonnegligible economic costs. That sequence-dependent biases extend well beyond gambling is demonstrated by Chen, Moskowitz, and Shue (2016), who document consistent negative autocorrelation across three institutionalised high-stakes settings — refugee asylum decisions, loan application reviews, and baseball umpire pitch calls — with more than 5% of decisions reversed or erroneous due to negative autocorrelation in some samples, ruling out case quality as an alternative explanation. Together these studies establish that sequence-dependent biases are economically meaningful, but the field setting precludes isolating the sequence effect from confounding environmental features or benchmarking welfare losses against individual-level optima. Croson and Sundali (2005) themselves anticipated this gap, acknowledging that the hot-hand pattern in casino data may partly reflect income or house money effects rather than genuine belief distortion, and calling for lab studies that manipulate win/loss history while keeping income constant and elicit beliefs directly.

Measuring these biases in a controlled setting introduces its own challenges. The dominant laboratory approach in prediction tasks presents participants with a terminal streak of length k and records either a binary side choice or a numerical probability rating (Croson & Sundali, 2005; Kong et al., 2020; Rao & Hastie, 2023; Xiang et al., 2025) — reducing a rich sequence to a single scalar summary and discarding everything that preceded the terminal run. This is consequential: which fallacy a given sequence elicits depends not only on the length of the terminal streak but on the full sequence structure — its overall switchiness or streakiness — that a terminal-streak design cannot detect (Asparouhova et al., 2009; Dorst, 2026; Xiang et al., 2025). Which fallacy a given sequence elicits is therefore an empirical question, not a property that can be read off from sequence properties alone — yet the dominant design forecloses

exactly that question by construction.

The present study is designed to fill the economic gap and sidestep the measurement problem. To close the economic gap, we present all 64 possible six-flip coin histories to participants in a controlled setting, fixing the generator as a known fair coin to isolate the effect of sequence structure, and elicit individual risk preferences via the Holt–Laury task (Holt & Laury, 2002). This allows us to compute certainty-equivalent welfare losses relative to each participant’s own optimum and decompose them across two margins — the extensive margin of whether to bet at all, and the intensive margin of how much to stake. To sidestep the classification problem, we propose an anchor-based approach: the maximally homogeneous pure sequences HHHHHH and TTTTTT serve as behavioral anchors, where betting with the streak unambiguously reveals hot-hand reasoning and betting against it unambiguously reveals gambler’s reasoning. Rather than imposing ex-ante predictions about which sequences are gamblerish or hot-handish based on their streakiness and switchiness, our approach infers the directional content of every sequence from the data. Together, these features allow us to provide the first jointly psychological and economic account of sequence-dependent betting behavior. The answer, as we show, is that the two dimensions are largely orthogonal — a finding that neither the field nor the prior lab literature was positioned to detect.

2 Methods

The experimental design, research questions, and empirical strategy were preregistered prior to data collection² at osf.io (DOI: 10.17605/OSF.IO/45NMK). The full replication package containing the experiment source code, data, and analysis scripts is available in the Data Availability section.

Participants completed an incentive-compatible sequence-betting task in which they observed each of the 64 possible six-toss coin histories (e.g., HHHHHH, THHHHH) exactly once and, for each, decided whether to bet on the next toss, which side, and how much to stake from a fresh 100-ECU endowment. After the betting task, participants completed a Holt–Laury risk elicitation task, whose estimates are used to construct individually optimal benchmarks under expected utility theory (EUT) against which betting behavior is evaluated. The payout multiplier was varied between subjects. In the *False-Negative* (FN) treatment ($m = 2.5$), EUT prescribes betting for all but extremely risk-averse participants; the primary deviation is therefore *under-betting*—failing to bet when betting is optimal. In the *False-Positive* (FP) treatment ($m = 1.9$), EUT prescribes abstention for all but extremely risk-loving participants; the primary deviation is *over-betting*—betting when abstention is optimal. The within-subject sequence design additionally enables analysis of side-choice behavior—whether participants systematically prefer Heads or Tails following particular histories—providing a direct behavioral test of the gambler’s fallacy and hot-hand fallacy at the sequence- and individual-level. Together, the design yields evidence on whether side-choice is biased, what economic consequences that bias entails, and which margin—extensive (whether to bet) or intensive (how much to stake)—drives any resulting welfare loss.

This within-subject design generates crossed random effects—each participant observes all 64 sequences—that are naturally accommodated by a hierarchical modelling approach. Frequentist alternatives would require either aggregating across participants or sequences, losing the granularity that motivates the design, or fitting fixed effects for all 64 sequences and participants separately, producing unstable estimates in the many cells with limited observations. All analyses are therefore conducted within a unified Bayesian hierarchical framework, which pools information across sequences and participants simultaneously, provides calibrated uncertainty estimates for sequence-level effects, and propagates uncertainty in the individually estimated risk parameters through to the welfare-loss

² [Note on pilot addendum](#)

calculations. Rather than testing whether effects differ from zero, inference asks whether posterior mass exceeds practically meaningful thresholds—a question the Bayesian framework answers directly and that aligns naturally with the preregistered design.

2.1 Experimental Design

In each of 64 rounds, participants observe a *sequence*—a six-toss coin history (e.g., HHHHHH, THHHHH)—and decide whether to bet on the next toss, which side (Heads or Tails) to bet on, and how much to stake from a fresh 100-ECU (Experimental Currency Unit) endowment. The 64 rounds present all possible sequences exactly once;³ participants are informed that each represents a hypothetical realization of a fair coin, not an outcome carried over from a previous round, but are not told that the full sequence space is exhausted. Sequences are presented in a randomized order, partitioned into four blocks of 16 with a mandatory rest break between blocks.

In each round, the participant begins with a fresh endowment of $e = 100$ experimental currency units (ECU). A successful bet of size b yields $e - b + mb$; an unsuccessful bet yields $e - b$; abstention (*No bet*) yields e , where m is the treatment-specific payout multiplier. The multipliers were chosen so that, under EUT with constant relative risk aversion (CRRA), the FN multiplier ($m = 2.5$) makes a strictly positive stake optimal for all but extremely risk-averse participants, while the FP multiplier ($m = 1.9$) makes full abstention optimal for all but extremely risk-loving participants. Coincidentally, the FP multiplier implies an expected value per unit staked comparable to an outside bet in American roulette with two zeros. Figure 1 shows the decision screen, comprising a progress bar, the displayed sequence, and a decision card with the three choice buttons (*Orel* = Tails, *Hlava* = Heads, *Nesázet* = No bet), a stake slider, and a real-time payoff preview conditional on the selected option and stake. The positions of the side buttons were randomized across rounds to prevent systematic left–right response biases, while the no bet button remained in a fixed third position. Figures in Online Appendix E.1 show the decision card at different stages of the betting decision. Response time from screen onset to choice confirmation was recorded throughout the task.

Before the betting task, participants provided informed consent, completed standardised instructions, a comprehension quiz, and one practice round. After the 64 betting rounds, they completed a 20-item Holt–Laury multiple price list to elicit individual CRRA risk-preference parameters. Each row of the price list offered a choice between a safe lottery (16 or 20 ECU with equal probability) and a risky lottery (1 or 38.5 ECU with equal probability), with the probability of the high payoff increasing from 0.05 to 0.95 in steps of 0.05 across rows. Participants then completed the Life Orientation Test–Revised (LOT-R) (Scheier et al., 1994), professionally back-translated into Czech, to measure dispositional optimism, followed by a short debriefing questionnaire. The full transcription of the experiment is available in the Czech original and English translation in Online Appendix F. The LOT-R original and translation are available in Online Appendix G.

2.2 General Modelling Conventions

All analyses are conducted within a unified Bayesian hierarchical framework estimated in Stan via the RStan interface (Stan Development Team, 2024) using dynamic Hamiltonian Monte Carlo. All models use four chains and 1,000 warmup draws. Convergence requires $\hat{R} \leq 1.01$ — stricter than the conventional 1.05 cutoff — an effective sample size ≥ 400 , and zero divergent transitions; these criteria

³ The exhaustive design was chosen to map the full sequence space and identify which specific histories drive belief distortions. Randomly realized histories would yield an unbalanced stimulus set—some sequences appearing multiple times, others not at all—making sequence-level inference noisy and precluding identification of the sequences most consequential for behavior. Waiting for all 64 unique sequences to occur naturally would also make session length impractically long and variable across participants.



Figure 1: Decision screen presented in each round of the experiment, FN treatment ($m = 2.5$; original interface in Czech). The progress bar at the top indicates the participant's position within the current block and overall session. Below, six cards show the current sequence of coin outcomes, ordered from oldest (left) to most recent (right); H denotes Heads (*Hlava*) and O denotes Tails (*Orel*). The sequence shown is 000H00. The header reads: "Your decision for the toss following displayed sequence." Participants choose among three options—*Orel* (Tails, selected and highlighted in blue), *Hlava* (Heads), or *Nesázet* (No bet)—and set their stake using a slider (1–100 ECU) accompanied by the prompt "Choose the stake size by clicking the slider (1–100 ECU)" The selected stake and side are summarised as "Stake: 25 ECU / Heads." The interface dynamically displays the implied payoffs: "If this round is selected for payment and the chosen side lands, you receive 138 ECU. If the other side lands, you receive 75 ECU." Participants confirm their decision by clicking *Pokračovat* (Continue).

are as preregistered. Where initial runs did not meet these criteria, sampler settings (`adapt_delta`, `max_treedepth`) and the number of post-warmup draws were increased until convergence was achieved: these adjustments affect only numerical sampling performance and do not alter likelihoods, priors, estimands, or reporting rules. The final sampler configuration for each model is reported in Online Appendix Table D1. Where relevant, confirmatory models RQ1–4 include an additive drift term to accommodate potential within-session changes such as learning or fatigue across the four experimental blocks,⁴ and a control for button order (Heads–Tails vs. Tails–Heads) to guard against positional response biases in side choice. Drift decisions for each confirmatory model are reported in Online Appendix Table D2; the button-order check is reported in Section 3.3. Unless stated otherwise, weakly informative priors are used throughout: $\mathcal{N}(0,1)$ or $\mathcal{N}(0,1.5)$ for location parameters and

⁴ Drift is assessed via within-participant permutation tests ($B = 2,000$) on block-level averages of the relevant outcome. Linear drift is included when the block-1 to block-4 difference Δ_{41} is significant at $\alpha = 0.05$ (two-sided); categorical drift when only the overall block range is significant (one-sided, $\alpha = 0.05$) but no monotone trend is detected; otherwise no drift term is added. Categorical drift uses block indicators $\gamma_{\text{block},k} \sim \mathcal{N}(0, 0.3)$ subject to $\sum_k \gamma_{\text{block},k} = 0$; linear drift uses the centered block index $\widetilde{\text{block}} \in \{-1.5, -0.5, 0.5, 1.5\}$ with $\gamma_{\text{drift}} \sim \mathcal{N}(0, 0.3)$. The preregistration specifies drift inclusion when diagnostics indicate systematic temporal patterns, without prescribing the specific test; the permutation-based operationalization was finalized during implementation and does not alter any inferential criteria or estimands.

HalfNormal(0, 1) for scale parameters, with model-specific deviations noted where they occur.

2.3 Modelling Strategy — Economic Consequences of Betting

This section addresses three questions concerning the economic consequences of sequence-dependent betting: whether sequence history affects the likelihood of betting (RQ1, extensive margin), the size of the stake conditional on betting (RQ2, intensive margin), and the resulting certainty-equivalent welfare loss (RQ3, economic consequence). All three are evaluated against each participant’s expected-utility benchmark, constructed from individually elicited risk preferences, and extended by an exploratory analysis of how betting and staking compare across treatments.

2.3.1 Expected-Utility Benchmark

Each participant i is assigned a CRRA risk-aversion parameter r_i , estimated from the Holt–Laury task using priors from Bland (2023); the model is estimated in Stan following the sampler configuration and convergence criteria of Section 2.2. Estimation details are documented in Online Appendix A.1. For a stake a from endowment $e = 100$ ECU under payout multiplier m , expected utility is

$$EU(a; r) = \frac{1}{2}u(e - a + ma; r) + \frac{1}{2}u(e - a; r), \quad u(x; r) = \begin{cases} \frac{x^{1-r}}{1-r}, & r \neq 1, \\ \ln x, & r = 1. \end{cases} \quad (1)$$

The EU-optimal stake $a^*(r_i)$ is the integer maximizer of $EU(a; r_i)$ over $\{0, 1, \dots, e\}$, where $a = 0$ denotes abstention. The two payout multipliers were chosen so that this optimum is informative in opposite directions: under the FN multiplier ($m = 2.5$) a strictly positive stake is optimal for all but extremely risk-averse participants, while under the FP multiplier ($m = 1.9$) full abstention is optimal for all but extremely risk-loving participants. To express deviations from this optimum in money-metric terms, we use the certainty equivalent $c(a; r)$ —the sure payoff equally preferred to the bet, defined by $u(c(a; r); r) = EU(a; r)$. The welfare loss from a participant’s observed stake a_{is} relative to their own optimum is

$$\Delta c_{is} = c(a_{is}^*; r_i) - c(a_{is}; r_i) \geq 0, \quad (2)$$

non-negative by construction since a_{is}^* maximizes expected utility and $c(\cdot; r)$ is strictly increasing in it. The benchmark a_{is}^* underlies the RQ2 stake-deviation estimand; Δc_{is} underlies the RQ3 welfare-loss estimand. Because r_i is estimated with posterior uncertainty, a^* , $c(\cdot)$, and Δc_{is} are computed draw-wise from the Holt–Laury posterior and summarized only after this propagation, rather than from a single point estimate of r_i . Risk parameters are estimated independently of the betting task itself, so r_i enters RQ1–3 as a fixed, externally measured benchmark against which betting is evaluated, not one that betting behavior can revise. Full details, including numerical safeguards, are in Online Appendix A.2.

2.3.2 Sample

RQs 1–3 are evaluated on a *confirmatory sample* comprising participants who satisfy three criteria: (i) assignment to the FN treatment, (ii) classification as a *normative better*, and (iii) consistent Holt–Laury risk preferences.

Regarding (i): the FN treatment is used exclusively because, under the multiplier $m = 2.5$, all but extremely risk-averse participants should normatively bet under expected utility theory. By contrast, under the FP multiplier $m = 1.9$, all but extremely risk-loving participants should normatively refrain from betting, making the normative benchmark less suitable for confirmatory inference.

Regarding (ii): to verify the normative assumption at the individual level, participants are formally classified as *normative betters* or *normative non-betters*. For each participant i and multiplier m , define

$$P_i^0(m) = P(a^*(r_i; m) = 0 \mid \text{HL data}), \quad P_i^+(m) = 1 - P_i^0(m). \quad (3)$$

In the FN treatment, participants with $P_i^0(2.5) \geq 0.90$ are classified as *normative non-betters* and excluded from the confirmatory sample; all remaining FN participants are, by construction, *normative betters*. (For completeness, in the FP treatment the classification is reversed: participants with $P_i^+(1.9) \geq 0.90$ are the *normative non-betters*.)

Regarding (iii): the confirmatory sample is further restricted to participants with consistent Holt–Laury preferences — a deviation from the original protocol motivated by the data and detailed in Section 3.1.

As a robustness check, RQs 1–3 are re-estimated on the *full FN sample* — all FN participants, without the normative-better or Holt–Laury-consistency restrictions. The *full FP sample*, used in the exploratory FN–FP contrasts, is defined analogously.⁵

2.3.3 Extensive Margin (RQ1)

RQ1 asks whether any sequences systematically suppress betting relative to the expected-utility–optimal benchmark, which prescribes universal betting in the confirmatory sample. Let $b_{is} \in \{0, 1\}$ denote whether participant i places a bet on sequence s . We fit a Bernoulli hierarchical logistic regression⁶ with crossed random effects for participants and sequences,

$$b_{is} \sim \text{Bernoulli}(\pi_{is}^b), \quad \text{logit}(\pi_{is}^b) = \alpha + u_i + \beta_s, \quad u_i \sim N(0, \sigma_u), \quad \beta_s \sim N(0, \sigma_s), \quad (4)$$

with sequence effects identified relative to the grand mean via $\sum_s \beta_s = 0$, so α represents the population-average log-odds of betting, and weakly informative priors $\alpha \sim N(0, 1.5)$, $\sigma_u, \sigma_s \sim \text{HalfNormal}(0, 1)$. The primary estimand is the sequence-level mean betting probability $\mu_s^b = \mathbb{E}_i[\pi_{is}^b]$, evaluated against a benchmark of universal betting ($\mu_s^b = 1$); posterior concentration below this benchmark is summarized by

$$U_s^b(\cdot) = P(\mu_s^b < 1 - \rho), \quad \rho = 0.10, \quad (5)$$

with sensitivity analyses at $\rho \in \{0.08, 0.12\}$. The model also yields participant-level mean betting probabilities $\mu_i^b = \mathbb{E}_s[\mathbb{E}(\pi_{is}^b \mid \text{data})]$, used to characterize individual underbetting tendencies.

2.3.4 Intensive Margin (RQ2)

RQ2 asks whether any sequences lead to over- or under-staking relative to the expected-utility–optimal amount. Let $\Delta a_{is} = a_{is} - a_{is}^*$ denote the deviation of the observed stake from the EU-optimal stake. To isolate sequence-level effects from individual differences in staking scale, deviations are standardized within participant,

$$Z_{is}^{\Delta a} = \frac{\Delta a_{is} - \overline{\Delta a_i}}{s_i^*(\Delta a)}, \quad s_i^*(\Delta a) = \max(\text{sd}_i(\Delta a), 2), \quad (6)$$

⁵ RQ2 and RQ3 are conditional on betting trials to avoid degenerate likelihood contributions; RQ2 additionally requires at least three such trials per participant, since fewer provide insufficient within-participant variation to construct $Z_{is}^{\Delta a}$. Both restrictions apply to all samples.

⁶ If posterior predictive checks indicate overdispersion in sequence-wise betting frequencies, the Bernoulli likelihood is replaced by a Beta-Binomial specification on aggregated sequence-level counts with identical priors and hierarchical structure.

and a Gaussian hierarchical regression⁷ with crossed random effects is fit on $Z_{is}^{\Delta a}$,

$$Z_{is}^{\Delta a} = \alpha + u_i + \beta_s + \varepsilon_{is}, \quad u_i \sim N(0, \sigma_u), \quad \beta_s \sim N(0, \sigma_s), \quad \varepsilon_{is} \sim N(0, \sigma), \quad (7)$$

with sequence effects identified relative to the grand mean via $\sum_s \beta_s = 0$ and weakly informative priors $\alpha \sim N(0, 1)$, $\sigma_u, \sigma_s, \sigma \sim \text{HalfNormal}(0, 1)$. The sequence-level estimand is recovered by mapping the fitted linear predictor back onto the ECU scale,

$$\mu_s^a = \frac{1}{e} \mathbb{E}_i \left[\overline{\Delta a_i} + (\alpha + u_i + \beta_s) s_i^*(\Delta a) \right], \quad (8)$$

and posterior concentration outside a practical-equivalence band is summarized by

$$U_s^a(\cdot) = P(\mu_s^a < -\rho), \quad O_s^a(\cdot) = P(\mu_s^a > \rho), \quad \rho = 0.05, \quad (9)$$

with sensitivity analyses at $\rho \in \{0.03, 0.08\}$. While the absolute level of μ_s^a provides a descriptive summary of aggregate calibration, interpretation primarily focuses on between-sequence differences – captured by the sequence effects β_s – since global calibration shifts may reflect imperfect transport of the Holt–Laury risk parameter to the betting task rather than sequence-specific distortions.⁸ The model also yields participant-level mean stake deviations $\mu_i^a = \mathbb{E}_s[\mathbb{E}(\Delta a_{is}/e \mid \text{data})]$, used to characterize individual over- or under-staking tendencies.

2.3.5 Welfare Loss (RQ3)

RQ3 asks whether any sequences increase certainty-equivalent losses from suboptimal betting and staking. The welfare loss $\Delta c_{is} = c(a_{is}^*; r_i) - c(a_{is}; r_i)$ is non-negative by construction (Section 2.3.1); values at or below zero, arising only from numerical noise near the optimum, are set to zero prior to estimation. We fit a hurdle–Gamma model⁹ with crossed random effects in both components,

$$\Delta c_{is} \sim \begin{cases} 0, & \text{prob. } \pi_{is}^0, \\ \text{Gamma}(k_{is}, \theta_{is}), & \text{prob. } 1 - \pi_{is}^0, \end{cases} \quad (10)$$

$$\text{logit}(\pi_{is}^0) = \alpha^{(0)} + u_i^{(0)} + \beta_s^{(0)}, \quad \log \mathbb{E}[\Delta c_{is} \mid \Delta c_{is} > 0] = \alpha^{(+)} + u_i^{(+)} + \beta_s^{(+)}, \quad (11)$$

with sequence effects identified relative to the grand mean via $\sum_s \beta_s^{(0)} = \sum_s \beta_s^{(+)} = 0$ and weakly informative priors $\alpha^{(0)}, \alpha^{(+)} \sim N(0, 1)$, $\sigma_u^{(0)}, \sigma_s^{(0)}, \sigma_u^{(+)}, \sigma_s^{(+)} \sim \text{HalfNormal}(0, 1)$, and $k \sim \text{Gamma}(2, 0.1)$ for the shape parameter of the Gamma component. The sequence-level estimand combines both behavioral margins into a single money-metric measure,

$$\mu_s^c = \mathbb{E}_i[\Delta c_{is}/e], \quad (12)$$

expressed as a share of the endowment, and posterior concentration above a practical-equivalence threshold is summarized by

$$L_s(\cdot) = P(\mu_s^c > \rho), \quad \rho = 0.05, \quad (13)$$

⁷ If posterior predictive checks detect boundary inflation caused by all-in choices ($a = 100$), a one-inflated Beta-Binomial model on the raw stake scale is estimated using the same hierarchical structure. Robustness checks additionally repeat the analysis using alternative floors $s_i^*(\Delta a) = \max(\text{sd}_i(\Delta a), c)$ with $c \in \{1, 5\}$ and a median absolute deviation in place of $\text{sd}_i(\Delta a)$.

⁸ This interpretive refinement was implemented in response to pilot data and is documented in the preregistration’s post-pilot addendum (Section C.4).

⁹ If the mean hurdle probability $\bar{\pi}^0 < 0.05$, a Gamma-only specification is fitted as the primary alternative. A Gaussian working model with the same hierarchical structure is additionally fitted as a diagnostic for the sign and relative ordering of sequence effects.

with sensitivity analyses at $\rho \in \{0.03, 0.08\}$. The model also yields participant-level welfare losses $\mu_i^c = \mathbb{E}_s[\mathbb{E}(\Delta c_{is}/e \mid \text{data})]$, used to characterize individual decision efficiency.

2.3.6 FN–FP Contrasts (EX3)

As an exploratory complement to the confirmatory analyses, RQs 1–3 are re-estimated on the full FP sample and contrasted with the full FN sample (see Section 2.3.2 for definition). These re-estimations additionally serve as a robustness check on the confirmatory results, since the full FN sample includes Holt–Laury-inconsistent participants excluded from the confirmatory sample. For each quantity $\mu \in \{\mu_s^b, \mu_s^a, \mu_s^c\}$, the sequence-level FN–FP contrast is defined as $\Delta_s = \mu_s^{\text{FN}} - \mu_s^{\text{FP}}$, with posterior medians, 95% credible intervals, and $P(\Delta_s > 0)$ reported. Because treatment is assigned between subjects, participant-level contrasts are defined at the group level as differences in treatment-level averages of participant-level summaries. For μ_s^a (RQ2), the FN–FP contrast reflects a mechanical consequence of the near-zero EU-optimal stake in the FP treatment rather than a substantive difference in staking behavior. Descriptive FN–FP contrasts are additionally reported for side-choice behavior (RQ4, Section 2.4.2), following the same conventions. All FP models use the same likelihoods, priors, drift adjustments, and diagnostic procedures as the corresponding FN models. These analyses are descriptive and do not introduce additional inferential criteria.

2.4 Modelling Strategy — Psychology of Sequence-Dependent Betting

2.4.1 Sample

The confirmatory analysis in this section (RQ4) is estimated on the full FN sample (Section 2.3.2). Unlike RQs 1–3, the normative-better and Holt–Laury-consistency restrictions do not apply, since side choice carries no normative prescription under a fair coin. RQ4 and EX1 are additionally restricted to betting trials; participants who placed no bets across all 64 sequences contribute no side-choice observations and are excluded by construction. The exploratory analyses (EX1, EX4) are estimated on the full FN sample; EX2 is estimated separately on the full FN and FP samples. Descriptive FN–FP contrasts for RQ4 are reported following the conventions of Section 2.3.6.

2.4.2 Directional Side Choice (RQ4)

RQ4 asks whether any sequences systematically bias side choices toward Heads or Tails. Let $h_{is} \in \{0, 1\}$ denote whether participant i chose Heads (1) or Tails (0) on sequence s . We fit a Bernoulli hierarchical logistic regression¹⁰ with crossed random effects for participants and sequences,

$$h_{is} \sim \text{Bernoulli}(\pi_{is}^h), \quad \text{logit}(\pi_{is}^h) = \alpha + u_i + \beta_s, \quad u_i \sim N(0, \sigma_u), \quad \beta_s \sim N(0, \sigma_s), \quad (14)$$

with sequence effects identified relative to the grand mean via $\sum_s \beta_s = 0$ and weakly informative priors $\alpha \sim N(0, 1.5)$, $\sigma_u, \sigma_s \sim \text{HalfNormal}(0, 1)$. The population baseline Heads probability is defined as

$$\bar{h} = \mathbb{E}_{u \sim N(0, \sigma_u)}[\text{logistic}(\alpha + u)], \quad (15)$$

integrating over participant heterogeneity, and the sequence-level mean Heads probability is

$$\mu_s^h = \mathbb{E}_{u \sim N(0, \sigma_u)}[\text{logistic}(\alpha + u + \beta_s)]. \quad (16)$$

¹⁰ If posterior predictive checks indicate overdispersion in sequence-wise Heads-choice frequencies, the Bernoulli likelihood is replaced by a Beta-Binomial specification on aggregated sequence-level counts with identical priors and hierarchical structure.

Both $\bar{h}$ and μ_s^h are defined as population expectations with respect to the participant heterogeneity distribution, ensuring they are on the same scale. Posterior concentration of μ_s^h outside a practical tolerance band around $\bar{h}$ is summarized by

$$H_s(\cdot) = P(\mu_s^h > \bar{h} + \delta), \quad T_s(\cdot) = P(\mu_s^h < \bar{h} - \delta), \quad \delta = 0.05, \quad (17)$$

with sensitivity analyses at $\delta \in \{0.03, 0.08\}$. The model also yields participant-level mean Heads probabilities $\mu_i^h = \mathbb{E}_s[\mathbb{E}(\pi_{is}^h \mid \text{data})]$, used to characterize individual directional tendencies.

2.4.3 Gamblerish–Hot-handish Patterns (EX1)

EX1 extends RQ4 by examining whether sequences systematically generate streak-following (hot-handish) or streak-breaking (gamblerish) choices. Rather than imposing ex-ante assumptions about what constitutes a streak, EX1 infers directional tendencies empirically through distributional similarity to two behavioral anchors: the pure sequences $\bar{H} = \text{HHHHHH}$ and $\bar{T} = \text{TTTTTT}$, on which betting with the streak is unambiguously hot-handish and betting against it unambiguously gamblerish.

For each sequence s , the RQ4 model yields posterior draws of the population Heads-choice probability $\theta_s^{(k)} = \mu_s^{h(k)}$, $k = 1, \dots, K$. Posterior similarity of sequence s to anchor $a \in \{\bar{H}, \bar{T}, 0\}$ is defined as

$$d_a(s) = P(|\theta_s - \theta_a| < \epsilon \mid \text{data}) \approx \frac{1}{K} \sum_{k=1}^K \mathbf{1}\left\{|\theta_s^{(k)} - \theta_a^{(k)}| < \epsilon\right\}, \quad (18)$$

where θ_0 is the neutral baseline defined draw-wise as $\theta_0^{(k)} = \mathbb{E}_{u \sim N(0, \sigma_u^{(k)})}[\text{logistic}(\alpha^{(k)} + u)]$, and $\epsilon = 0.05$ with sensitivity analyses at $\epsilon \in \{0.03, 0.08\}$. Normalized similarity weights are

$$w_a(s) = \frac{d_a(s) + \eta}{d_{\bar{H}}(s) + d_{\bar{T}}(s) + d_0(s) + 3\eta}, \quad a \in \{\bar{H}, \bar{T}, 0\}, \quad (19)$$

with stabilizing constant $\eta = 10^{-6}$, so $w_{\bar{H}}(s) + w_{\bar{T}}(s) + w_0(s) = 1$. Similarity indicators and weights are computed within each posterior draw to propagate uncertainty, yielding draw-specific directional scores. For each betting trial (i, s) , the trial-level directional score is

$$z_{is} = (2h_{is} - 1)(w_{\bar{H}}(s) - w_{\bar{T}}(s)), \quad (20)$$

so $z_{is} \in [-1, 1]$ with values near +1 indicating hot-handish and near −1 gamblerish responding. Sequence- and participant-level directional tendencies are

$$\chi_s = \mathbb{E}_i[z_{is}], \quad \chi_i = \mathbb{E}_s[z_{is}], \quad (21)$$

where χ_i is the Gamblerish–Hot-handish Index (GHI).¹¹

2.4.4 Associations with Optimism and Response Time (EX2)

EX2 examines whether dispositional optimism (LOT-R) and mean response time predict participant-level behavioral outcomes across RQs 1–4. The standardized predictors are Z_i^{opt} (LOT-R score standardized across both treatments) and Z_i^{rt} (standardized log mean response time). The four participant-level outcomes are taken from the posterior estimates of the confirmatory models: μ_i^b (betting probability, RQ1), μ_i^a (proportional stake deviation, RQ2), μ_i^c (certainty-equivalent welfare

¹¹ The preregistered analysis plan additionally specified predictors of the GHI and a pure-sequence approximation to the full index; these are not reported in the main text as the anchor-based framework collapsed due to the pure sequences turning out behaviorally neutral (see Section 3.3 and Online Appendix B.2 for details).

loss, RQ3), and μ_i^h (baseline Heads probability, RQ4). Each outcome is standardized within each posterior draw across participants before entering the regression. A multivariate Bayesian regression with partial pooling across outcomes is estimated separately for each treatment. Let y_i^k denote the standardized posterior draw for outcome $k \in \{b, a, c, h\}$ of participant i . The model is

$$y_i^k = \alpha_k + \beta_{k,\text{opt}} Z_i^{\text{opt}} + \beta_{k,\text{rt}} Z_i^{\text{rt}} + \varepsilon_i^k, \quad \varepsilon_i^k \sim N(0, \sigma_k), \quad (22)$$

with slopes partially pooled across outcomes via a non-centred parameterisation,

$$\beta_{k,\text{opt}} = \bar{\beta}_{\text{opt}} + \tau_{\text{opt}} \cdot \tilde{\beta}_{k,\text{opt}}, \quad \beta_{k,\text{rt}} = \bar{\beta}_{\text{rt}} + \tau_{\text{rt}} \cdot \tilde{\beta}_{k,\text{rt}}, \quad (23)$$

and weakly informative priors $\bar{\beta}_{\text{opt}}, \bar{\beta}_{\text{rt}} \sim N(0, 1)$, $\tau_{\text{opt}}, \tau_{\text{rt}} \sim \text{HalfNormal}(0, 0.5)$, $\tilde{\beta}_{k,\text{opt}}, \tilde{\beta}_{k,\text{rt}} \sim N(0, 1)$, $\alpha_k \sim N(0, 1)$, $\sigma_k \sim \text{HalfNormal}(0, 1)$. Posterior uncertainty in the outcomes is propagated via a multiple-imputation likelihood that marginalises over upstream posterior draws within a single model fit. Posterior medians, 95% credible intervals, and $P(\cdot > 0)$ are reported for the pooled slopes $\bar{\beta}_{\text{opt}}, \bar{\beta}_{\text{rt}}$ and the outcome-specific slopes $\beta_{k,\text{opt}}, \beta_{k,\text{rt}}$. Analyses involving μ_i^a are restricted to participants with at least three betting trials (Section 2.3.2).

2.4.5 Canonical Rule-Based Classifications (EX4)

EX4 examines how closely canonical rule-based classifications from the hot-hand and gambler's fallacy literature correspond to the sequence-level directional estimates from RQ4.¹² For each betting trial (i, s) and canonical rule k , the observed side choice is classified as hot-handish (+1) or gamblerish (−1), yielding indicators $q_{is}^{(k)} \in \{-1, +1\}$; trials for which a rule is undefined are omitted. Five canonical rules are considered, defined in terms of the six-toss sequence $(x_{s1}, \dots, x_{s6})$ with $x_{st} \in \{+1, -1\}$ corresponding to Heads and Tails, and participant i 's chosen side $c_{is} \in \{+1, -1\}$: (i) *Last-outcome rule*: $q_{is}^{(\text{last})} = c_{is} \cdot x_{s6}$. (ii) *Run-length rule* ($\ell_s \geq 2$): $q_{is}^{(\text{run2})} = c_{is} \cdot x_{s6}$ if the terminal run length $\ell_s \geq 2$, undefined otherwise. (iii) *Run-length rule* ($\ell_s \geq 3$): $q_{is}^{(\text{run3})} = c_{is} \cdot x_{s6}$ if $\ell_s \geq 3$, undefined otherwise. (iv) *Imbalance rule*: $q_{is}^{(\text{imb})} = -c_{is} \cdot \text{sign}(B_s)$, where $B_s = \sum_{t=1}^6 x_{st}$ is sequence imbalance; sequences with $B_s = 0$ are unclassified. (v) *Alternation rule*: $q_{is}^{(\text{alt})} = c_{is} \cdot x_{s6}$ if $A_s \leq 2$, $-c_{is} \cdot x_{s6}$ if $A_s \geq 4$, undefined if $A_s = 3$, where $A_s = \sum_{t=2}^6 \mathbf{1}\{x_{st} \neq x_{s,t-1}\}$ is the number. For each rule k , trial-level classifications are aggregated across participants to form sequence-level scores $Q_s^{(k)} = N_s^{(k)-1} \sum_i q_{is}^{(k)}$, where $N_s^{(k)}$ is the number of included betting observations for sequence s under rule k . Association with the RQ4 sequence-level estimates is summarized by the Spearman correlation between $Q_s^{(k)}$ and the posterior median of μ_s^h across the 64 sequences, with p -values reported for each rule k . Correlations are additionally reported against the posterior probabilities of Head- and Tail-bias, $H_s(\cdot)$ and $T_s(\cdot)$, from RQ4.

2.5 Descriptive Classification Labels

Each confirmatory model yields posterior distributions for sequence- and participant-level quantities. For interpretive convenience, sequences are assigned descriptive labels – *strong*, *moderate*, *weak*, or *neutral* – reflecting the magnitude of posterior concentration outside the practical equivalence region in the relevant direction (e.g., strong underbetting, moderate Head-bias). Participants are analogously labeled – *solid*, *likely*, *leaning*, or *neutral* – reflecting the strength of their individual tendency (e.g., solid under-staker, leaning welfare loser). These labels are purely expository and do not function as inferential criteria; all

¹² The preregistered specification compared canonical rules to the anchor-based directional tendencies χ_s from EX1 with full posterior uncertainty propagation; this was not estimable due to the collapse of the EX1 framework. EX4 therefore proceeds against posterior medians of μ_s^h from RQ4 directly, without posterior propagation. The preregistered specification is described in Online Appendix B.2.3.

substantive conclusions rely on the full posterior distributions. The full labeling conventions, including the specific probability thresholds for each RQ, are detailed in Online Appendix B.1.

2.6 Participants and Procedure

Participants were recruited from the ORSEE (Greiner, 2015) subject pool maintained by the Laboratories of Behavioral Studies at Jan Evangelista Purkyně University in Ústí nad Labem, comprising approximately 1,500 registered individuals drawn primarily from the local student population, with women outnumbering men roughly two to one. Sessions were conducted on March 17 and 18, 2026, with participants seated at PC terminals according to a number drawn before entering the laboratory. Treatment was assigned randomly at session start following the preregistered 3:1 (FN:FP) allocation: given a subject pool expected to yield 200–250 participants, the larger share was allocated to the confirmatory FN treatment, where sequence-level estimation demands a larger sample, and the smaller share to the exploratory FP treatment, where betting was expected to be rare, so allocating a larger sample to this purely descriptive arm would have added little inferential value.

The main study yielded 202 participants (149 FN, 53 FP), generating 9,536 sequence-level observations in the FN treatment and 3,392 in the FP treatment. No participant was excluded. Compensation consisted of a show-up fee of 150 Czech Crowns (CZK) plus performance-based pay from two sources: one randomly selected round from the sequence-betting task and one randomly selected choice from the Holt-Laury task. Average total performance pay was roughly 125 ECU in both treatments, converted to CZK at a 1:1 ratio. In total, participants on average earned 275 CZK. For reference, this was roughly 11.5 euro or 13 U.S. dollars at the time of running the experiment.

The experiment was implemented in oTree (Chen, Schonger, & Wickens, 2016); all randomization (treatment assignment, sequence order, button order, coin tosses, and reward selection) used oTree's default pseudorandom number generator. Full oTree code is available within the replication package, see the Data Availability section. No deception was used and all procedures were approved by the Ethical Committee of the Faculty of Social and Economic Studies, Jan Evangelista Purkyně University in Ústí nad Labem.

Data collection was preceded by a pilot run conducted on January 28, 2026 (14 participants in three sessions, excluded from all analyses). The pilot was designed to verify software execution, test the decision interface, and validate prepared analysis scripts against dummy data. Based on the pilot findings, a preregistration addendum was filed specifying technical modeling details; no inferential criteria or research questions were modified. In each pilot session, four participants (12 in total) were additionally equipped with eye-tracking devices (Tobii 60 Hz) to verify that the decision interface functioned as intended. Eye-tracking was restricted to the pilot: as the laboratory operates only four units, deploying it in the main experiment would have required substantially smaller concurrent session sizes, imposing logistical costs disproportionate to the inferential value of the data. Eye-tracking data are not part of the preregistered outcomes; any descriptive use is limited to illustrative summaries of visual attention allocation.

3 Results

Section 3.1 provides descriptive statistics for the experimental sample. We report the distribution of individual risk-preference parameters estimated from the Holt–Laury task, summary statistics for betting decisions and stake choices from the sequence betting task, and distributions of two auxiliary measures: response time and dispositional optimism (LOT-R score). Section 3.2 examines the economic consequences of sequence-dependent betting. We analyze both the extensive margin (whether participants bet) and the intensive margin (how much they stake) and translate deviations from expected-utility-optimal behavior into certainty-equivalent welfare losses. The main analyses are conducted on the FN treatment, in which abstention from betting predominantly constitutes a false-negative error (the 2.5 multiplier on stakes in successful bets means that betting is expected-utility-optimal for all but extremely risk-averse participants). The FP treatment—in which betting predominantly constitutes a false-positive error (the 1.9 multiplier implies that abstention is expected-utility-optimal for all but extremely risk-loving participants)—serves as a descriptive contrast, with comparisons reported for the full samples of both treatments. The analyses in this section correspond to preregistered research questions 1–3 and exploratory analysis 3. Section 3.3 examines the psychological determinants of sequence-dependent side choices. We first test for unconditional directional biases toward Heads or Tails, then assess whether the shape of the observed sequence systematically shifts the direction of bets, connecting these patterns to the gambler’s fallacy and the hot-hand fallacy. We further examine whether dispositional optimism and response time predict individual tendencies toward streak-following or streak-reversing behavior, and assess the correspondence between the anchor-based directional measure developed in this study and canonical rule-based classifications from the existing literature. The analyses in this section correspond to preregistered research question 4 and exploratory analyses 1, 2, and 4. Finally, Section 3.4 ... **finish**

3.1 Descriptive Statistics

Table 1 reports descriptive statistics by treatment. In total, 202 participants completed the study, with a roughly 2:1 female-to-male ratio¹³ and mean age of 23.8 years. At the descriptive level, the two treatments are broadly comparable across all listed variables. Generally, the relatively high standard deviations across the board suggest substantial heterogeneity in individual behavior across tasks—a point we detail in the analytical part. The average risk aversion of $\bar{r} = 0.88$ ($s = 0.58$) is above levels typically reported in the literature, driven primarily by inconsistent Holt–Laury participants ($\bar{r} = 1.35$) relative to consistent ones ($\bar{r} = 0.64$). The latter is remarkably close to the classical switching-point estimate of $\bar{r} = 0.56$ ($s = 0.47$).¹⁴ As the inflated r_i estimates of inconsistent participants could compromise the EU-optimal stake and welfare loss computations on which analyses below rely, we elevate the preregistered robustness analysis on consistent participants to the primary analysis for RQ1–3.¹⁵ Online Appendix Figure E1 shows the pooled distribution of r_i within each treatment, which spans a wide range from near risk-neutral to strongly risk-averse preferences. (This spread later proves consequential for individual-level stake calibration in Section 3.2.) Part of this range is explained by sex: among consistent participants,

¹³ The imbalance is driven by the FN treatment, as the ratio is effectively balanced in the FP treatment. Since participants were randomly assigned to treatments in the prespecified 3:1 ratio, this is attributable to chance.

¹⁴ This observation provides reassurance that the Bayesian estimates are well-calibrated for participants with interpretable Holt–Laury responses. The consistent-participant mean of $\bar{r} = 0.64$ sits at the upper boundary of the range $-0.15 < r < 0.68$ that the Bland (2023) prior $r_i \sim \mathcal{N}(0.27, 0.36)$ is calibrated to cover for 75% of participants based on Holt and Laury (2002). This suggests the prior is somewhat conservative for this sample but not severely misspecified. As a robustness check, re-estimating r_i on the consistent subsample alone yields virtually identical estimates to those obtained by filtering consistent participants from the full-sample model.

¹⁵ Since the analyses focus on relative comparisons across sequences and participants rather than absolute magnitudes, the cost of retaining unreliable r_i estimates outweighs the cost of the reduced sample. This preregistration deviation is documented in Online Appendix H.

women in the FN treatment average $\bar{r} = 0.67$ ($n = 59$, 95% CI = [0.55, 0.80]) and men $\bar{r} = 0.57$ ($n = 35$, CI = [0.39, 0.75]); in the FP treatment, women average $\bar{r} = 0.78$ ($n = 20$, CI = [0.55, 1.01]) and men $\bar{r} = 0.54$ ($n = 16$, CI = [0.27, 0.80]). Online Appendix Figure E2 shows the posterior distribution of r_i by treatment and sex. In both treatments, women are on average more risk-averse than men, consistent with prior literature, though no group mean falls outside the confidence interval of any other group. The absence of a meaningful treatment difference within each sex further suggests that the Holt–Laury task was not systematically affected by prior exposure to different multipliers in the sequence-betting task.

	FN ($m = 2.5$)	FP ($m = 1.9$)
Sample Size	149	53
of which Women, Men, Unspecified	102, 43, 4	28, 25, 0
Tasks' Earnings in ECU	125 [62.6]	127 [47.0]
<i>Holt–Laury Task</i>		
$\bar{r}$ (Sample, Bayesian)	0.88 [0.58]	0.88 [0.57]
Share of Consistent Participants	66%	68%
$\bar{r}$ (Consistent Participants, Bayesian)	0.64 [0.51]	0.67 [0.53]
$\bar{r}$ (Consistent Participants, Switching Point)	0.56 [0.47]	0.57 [0.46]
<i>Sequence Betting Task (Full Sample)</i>		
Total Sequences	9,536	3,392
Share of Bets Placed	77% [42%]	68% [47%]
Normative Non-Bettors [†]	0	51
Normative Bettors [†]	149	2
Share of Endowment Staked in Bets Placed	49% [31%]	49% [31%]
Share of Heads in Bets Placed	51%	52%
<i>Additional Measures</i>		
Seconds to Decide per Sequence	7.3 [6.2]	6.7 [5.6]
Life Orientation Test Revised (0–24 pts.)	12.6 [4.7]	13.3 [4.3]
Life Orientation Test Revised (Cronbach's α)	0.80	

Table 1: Descriptive statistics by treatment. Entries are means with standard deviations in square brackets, or counts, or shares. [†]Model-derived classification based on whether the posterior probability of a strictly positive (normative betters) or zero (normative non-bettors) EU-optimal stake exceeds 90%.

The most striking descriptive observation concerns betting participation. In the FN treatment, all 149 participants are classified as *normative betters* — that is, each has at least a 90% posterior probability that the EU-optimal stake is positive (see details in Section 2.3.2). Betting should thus be universal, yet only 77% of decisions result in a bet. This 23-percentage-point gap likely reflects both residual uncertainty in the normative classification and the well-documented limits of EUT as a universal descriptive model. The situation in the FP treatment is harder to dismiss on the same grounds: with 51 of 53 participants classified as *normative non-bettors* (90% posterior probability that the EU-optimal stake is zero), abstention should be near-universal — yet two in three decisions still result in a bet. Figure 2, showing the distributions of raw participant-level betting rates (left panel) and of raw sequence-level stake amounts conditional on betting (right panel), provides further details. The left panel reveals that both distributions are left-skewed and concentrated above 50%, even though one would expect them to mirror each other given the opposing normative prescriptions. The key difference is that the FN treatment has a larger mass of often- to near-always-betting participants, suggesting that the favorable multiplier is more effective at encouraging betting than the unfavorable multiplier is at discouraging it. The right panel shows that the stake distributions are also remarkably similar across treatments, with both exhibiting a pronounced trimodal pattern concentrated at low stakes, 50 ECU, and the full endowment of 100 ECU. The only notable difference is that the unfavorable multiplier prompts somewhat fewer all-in bets and more small stakes than the favorable one. At the aggregate level, betting and staking patterns thus

appear largely unresponsive to the reward multiplier, a pattern that EUT alone cannot explain.

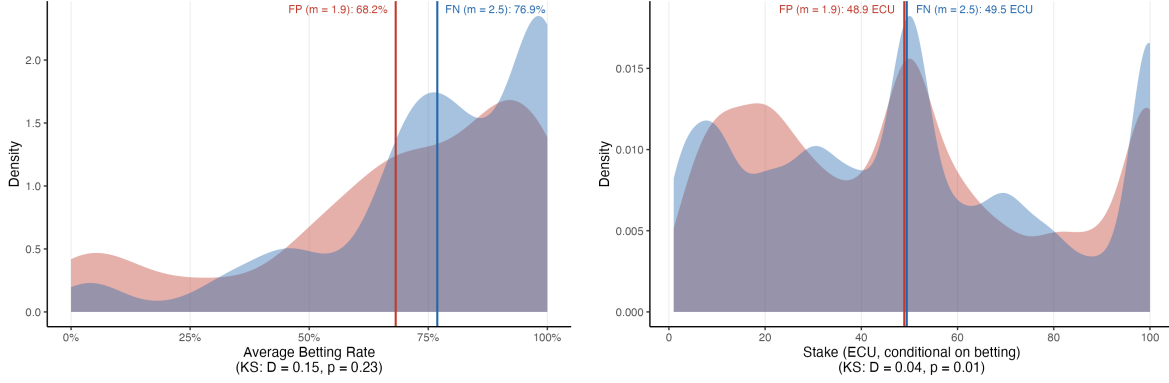


Figure 2: Distribution of individual betting behavior by treatment. *Left panel*: participant-level average raw betting rates across all 64 sequences (FN: $n = 149$; FP: $n = 53$). *Right panel*: sequence-level raw stake amounts conditional on betting, positive stakes only (FN: $n = 9,536$; FP: $n = 3,392$). Vertical lines mark treatment means. The near-identical distributions across treatments contrast sharply with the normative prescriptions: in FN, betting is EU-optimal for all 149 participants; in FP, abstention is EU-optimal for 51 of 53. Kolmogorov–Smirnov test statistics are shown in axis labels. For stakes, $D = 0.04$ indicates a negligible practical difference despite statistical significance ($p = 0.01$, approximate due to ties in the discrete stake variable), driven by the large number of sequence-level observations.

Additionally, we report two variables used in the preregistered exploratory analysis: sequence-level response times and participant-level dispositional optimism. Figure 2 shows that response time distributions are broadly similar across treatments, with the only visible difference being a slightly larger mass of fast responses (1–2 seconds) in the FP treatment, likely reflecting participants who abstain immediately without deliberation. LOT-R score means are similarly distributed between treatments, with FP participants showing a slightly greater concentration at the higher end of the optimism scale. This pattern could stem from the smaller FP sample or its more balanced sex ratio. As a complementary check on what participants attended to while making their decisions, we collected eye-tracking data in a pilot session, tracking fixations on five areas of interest. The large majority of visual attention was allocated to *Sequences* and *Payoff*, consistent with participants treating the outcome history and payoff information as the primary inputs to their betting decisions. Full details of the eye-tracking procedure and results are reported in Online Appendix C.9.

3.2 Economic Consequences of Sequence-Dependent Betting

The main analysis of economic consequences focuses on the FN treatment, where the expected-utility-optimal action is to place a strictly positive bet for all participants in our sample. Comparing the certainty equivalent of each observed stake to the maximum attainable under the individually optimal stake, we find that participants incurred a mean certainty-equivalent welfare loss of 12.5% (95% CI = [11.8%, 13.5%]) of the 100 ECU trial endowment. This aggregate figure, however, masks substantial heterogeneity: welfare losses reflect both the decision to bet and, conditional on betting, the size of the stake, both of which vary considerably across sequences and participants. We proceed in three steps, first establishing the individual-level effects against which sequence-level effects are interpreted (*Individual Effects*), then examining sequence-level effects themselves (*Sequence Effects*), before concluding with a full-sample between-treatment comparison (*Treatment Comparison*).

Individual Effects

Table 2 illustrates that individuals differ substantially in their baseline propensity to bet and in their staking behavior, such that sequence-level effects are best understood as modulations around highly

	Betting rate $\hat{\mu}^b$	Stake deviation $\hat{\mu}^a$	Welfare loss $\hat{\mu}^c$
EU-optimal benchmark	100%	0%	0%
Grand mean	74.2% [73.4%, 75.0%]	+7.3% [+6.9%, +7.8%]	12.5% [11.8%, 13.5%]
σ_u/σ_s	3.07/0.80 = 3.87	0.01/0.19 = 0.05 [†]	1.33/0.21 = 6.41
Classification (%)	solid under: 56% neutral: 43%	solid over: 50% solid under: 37% neutral: 12%	solid: 43% neutral: 52%
Likelihood	Bernoulli	Gaussian	Gamma
Participants	98	95 [‡]	98
Observations	6,272	4,653	6,272

Table 2: Model summaries for the three behavioral quantities, FN treatment, restricted to consistent Holt–Laury participants. Posterior medians with 95% credible intervals in brackets. σ_u and σ_s are the posterior median between-participant and between-sequence SDs, whose ratio reflects the relative contribution of individual heterogeneity versus sequence-level variation; values above 1 indicate that who you are matters more than which sequence you face. Classification percentages are based on posterior concentration outside the preregistered practical equivalence region. [†]The near-zero ratio for stake deviation reflects the within-participant standardization of stake deviations ($Z_{is}^{\Delta a}$): by construction, between-participant variation in the level of staking is absorbed by the standardization, leaving sequence effects as the dominant source of variation in the normalized outcome. [‡]RQ2 excludes three normative betters (EUT prescribes a positive stake given their risk preferences) who placed too few bets in practice to identify within-participant stake variation.

heterogeneous individual baselines. To further characterize individual heterogeneity, we classify participants as *solid*, *likely*, *leaning*, or *neutral underbetters*, *over-* or *under-stakers*, and *welfare losers*, based on posterior concentration outside the preregistered practical equivalence region for quantities $\hat{\mu}_i^b$, $\hat{\mu}_i^a$, and $\hat{\mu}_i^c$, respectively. Figure 3 presents the individual posterior distributions for all three quantities, with blue distributions showing the confirmatory subset. The gray ghost distributions show the full FN sample including both consistent and inconsistent participants. The figure reveals substantial heterogeneity across all three quantities. The top-left panel illustrates that while a large portion of participants bet on almost every sequence (43% are classified as neutral, implying near-optimal betting rates), an equally large portion bets selectively or rarely (56% fall well outside the equivalence region, classified as solid underbetters), with relatively few participants in between. The top-right panel (“the hat”) reveals that rather than a uniform shift toward over-staking (implied by the grand mean of +7.3%), the sample is characterized by two roughly equally sized groups pulling in opposite directions: 50% stake well above the EU-optimal amount (solid over-stakers) and 37% well below it (solid under-stakers), with only 12% staking close to optimal.¹⁶ The bottom panel reveals that the strongly right-skewed distribution qualifies the aggregate grand mean of 12.5%: 52% of participants incur negligible losses regardless of their betting frequency, while 43% fall well above the equivalence threshold (solid welfare losers). The welfare losers are almost exclusively frequent bettors whose losses arise from miscalibrated stake sizes rather than from abstaining from betting.

Sequence Effects

Sequence-level effects are estimated on top of the substantial individual heterogeneity documented above; nonetheless, all 64 sequences are classified as strongly underbetting and strongly welfare-losing. Stake calibration is the only dimension showing variation in strength, and even there all 64 sequences have credible intervals (CI) lying entirely above zero, with 36% of sequences classified as strong over-staking, 19% moderate, 33% weak, and 12% neutral. Online Appendix Figure E7 reports sequence-level posterior

¹⁶ This split is not fully independent of risk preference: among confirmatory-sample participants, r_i correlates strongly with the individual stake deviation $\hat{\mu}_i^a$ (Pearson $r = 0.57$, Spearman $\rho = 0.57$, $n = 95$, $p < 0.001$), and the two groups differ sharply in average risk aversion ($\bar{r} = 0.30$ for under-stakers vs. $\bar{r} = 0.84$ for over-stakers). Part of the bimodality in $\hat{\mu}_i^a$ is therefore a byproduct of how the EUT-optimal benchmark stake is constructed from r_i , rather than evidence of two independent behavioral types.

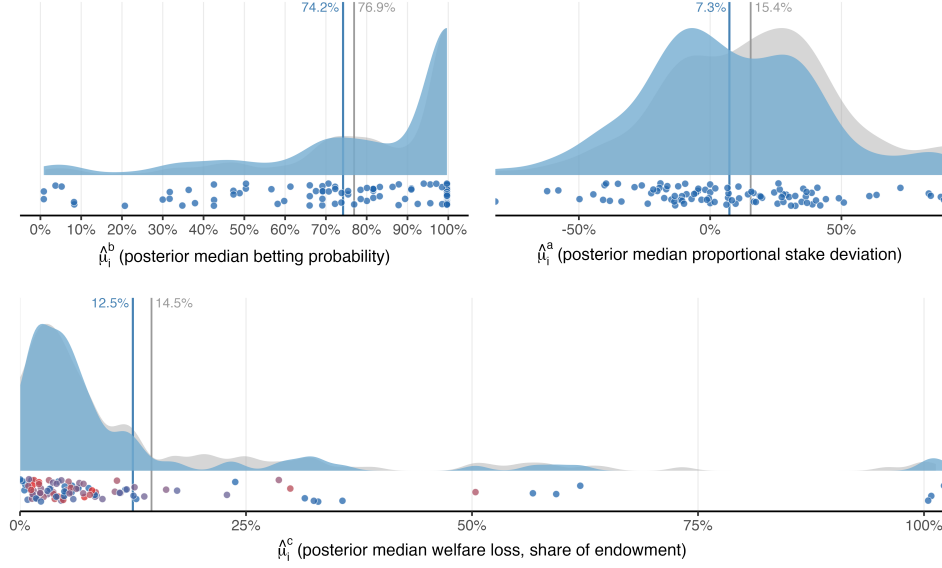


Figure 3: Distribution of individual-level posterior median estimates for the FN treatment. Blue distributions and points show the confirmatory subset (consistent Holt–Laury participants); grey ghost distributions show the full sample as a robustness reference. Solid vertical lines mark the confirmatory grand means (posterior medians); dashed lines mark the full-sample grand means. Each point represents one participant’s posterior median; shaded areas show kernel density estimates. *Top-left panel*: betting probability ($\hat{\mu}_i^b$; grand mean = 74.2%, full sample = 76.9%). The distribution is strongly right-skewed with a large mass near 100% and a smaller cluster of near-never-bettors. *Top-right panel*: proportional stake deviation ($\hat{\mu}_i^a$; grand mean = 7.3%, full sample = 15.4%). The broad distribution conceals two roughly equally sized subpopulations of solid over- and under-stakers whose opposing tendencies largely cancel at the aggregate level. The shape of the distribution is remarkably close to that of Saint-Exupéry’s *Drawing Number One* (Saint-Exupéry, 1943), which illustrates the idea that true contents are not immediately apparent from the overall shape. *Bottom panel*: certainty-equivalent welfare loss ($\hat{\mu}_i^c$; grand mean = 12.5%, full sample = 14.5%). Points are colored by median betting probability (red = low, blue = high). The distribution is strongly right-skewed; participants in the right tail are predominantly frequent bettors, indicating that high welfare losses arise from intensive-margin miscalibration rather than abstention alone.

means and 95% CIs for *Welfare loss*, *Betting rate*, and *Stake deviation*, one panel per quantity, with sequences ordered by posterior mean *Welfare loss*. Each panel also shows the posterior grand mean and its 95% CI to aid relative comparison across sequences. The pure sequences, HHHHHH and TTTTTT, are the most extreme across all three quantities. For *Welfare loss* and *Stake deviation*, this extremity holds both in absolute terms (measured against the normative benchmark) and in relative terms (measured against the other sequences). For *Betting rate*, they are extreme only in absolute terms: several other sequences elicit similarly sized relative over- or under-betting. Their welfare losses average 30% for HHHHHH and 26% for TTTTTT, considerably above the grand mean of 12.5% that they themselves help drive upward. The pure sequences were preregistered as behavioral anchors for the gambler’s- and hot-hand-fallacy analysis; whether their extremity on these economic margins also translates into a meaningful directional signal is examined in Section 3.3. Beyond these two outliers, the remaining 62 sequences reveal a more systematic pattern in how betting and staking relate to welfare losses, shown in Figure 4: it plots each sequence’s *Betting-rate* Δ and *Stake-deviation* Δ (posterior mean minus the sequence’s own grand mean) against the sequence’s rank in the *Welfare-loss* ordering.¹⁷ *Betting-rate* Δ traces a clear upward trend across the ranking, while *Stake-deviation* Δ stays scattered close to zero throughout, with no comparable trend.¹⁸ This indicates that sequences drive welfare losses primarily by pulling betting probabilities away from

¹⁷ The ordering of these 62 sequences is only weakly identified: individual heterogeneity dominates sequence-level variation in welfare loss ($\sigma_u/\sigma_s = 6.41$), and cross-model ordering stability is moderate (Spearman $\rho \approx 0.49$ – 0.57). The welfare-loss rank used here, and the trend fit on it, should accordingly be interpreted with caution beyond the two pure sequences.

¹⁸ Excluding the two pure sequences, *Betting-rate* Δ rises credibly with welfare-loss rank ($b = 0.0018$, $SE = 0.0005$, $p = .001$, $R^2 = .18$), while the *Stake-deviation* Δ slope, though nominally distinguishable from zero ($b = -0.0003$, $SE = 0.0001$, $p = .018$), is an order of magnitude smaller and explains far less variance ($R^2 = .09$).

the grand mean, rather than through any systematic miscalibration of stake size.¹⁹

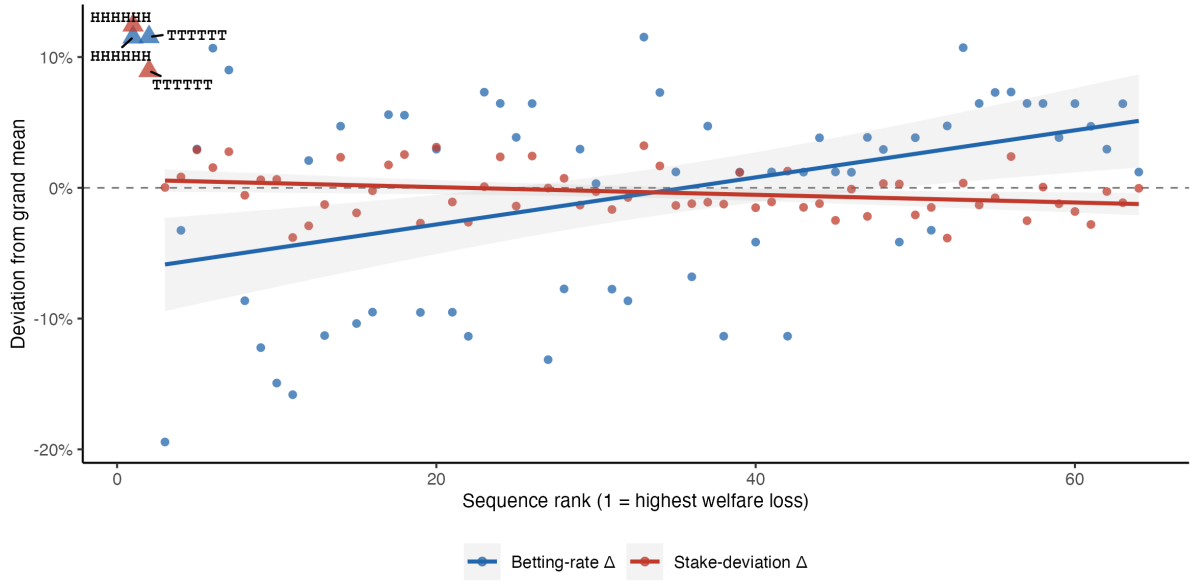


Figure 4: Sequence-level RQ1 (betting probability) and RQ2 (stake deviation) posterior means, each expressed as a deviation from its own grand mean, plotted against the sequence’s rank in the RQ3 welfare-loss ordering (rank 1 = highest loss). The two pure sequences (HHHHHH, TTTTTT) are marked with triangles and labelled; they are excluded from the fitted OLS trend lines (shaded bands: 95% CI) since both margins deviate sharply for these two sequences alone. Fit on the remaining 62 sequences, the betting-probability slope is credibly positive ($b = 0.0018$, $SE = 0.0005$, $p = .001$, $R^2 = .18$), while the stake-deviation slope, though nominally distinguishable from zero ($b = -0.0003$, $SE = 0.0001$, $p = .018$), is an order of magnitude smaller and explains far less variance ($R^2 = .09$).

Treatments Comparison

Although the FP treatment was preregistered as exploratory and assigned a smaller sample, on the expectation that its EU-optimal abstention benchmark would leave too few bets for a full hierarchical analysis, FP participants in fact bet at rates and stakes comparable to those observed in FN — a striking convergence given that the two frames carry opposite normative prescriptions. Despite this, modelled welfare loss is comparable across treatments (medians 14.5% FN vs. 15.7% FP; median $\Delta = -0.02$, 95% CI $[-0.03, -0.01]$), so FP participants do not appear to incur meaningfully larger welfare losses than their more “rational” FN counterparts. Betting rates differ only modestly between frames (76.9% vs. 68.2%), while the stake deviation differs mechanically, since the EU-optimal stake is near zero under FP. Full contrasts, sequence-level detail, and robustness checks are reported in Online Appendix C.1.

3.3 Psychology of Sequence-Dependent Betting

The analyses so far have addressed the economic dimension of betting behavior: whether participants bet, how much they stake, and at what welfare cost. We now turn to the psychological dimension. Setting aside whether and how much to bet, participants face a directional choice on every betting trial — Heads or Tails — that carries no expected payoff difference under a fair coin. Any systematic departure from indifference therefore reflects pure psychological bias rather than rational calculation. We examine this bias at two levels: the individual participant, characterising the distribution of baseline Heads tendencies across the population, and the sequence, asking whether the sequences shown to participants shifts their side choice in predictable ways. As preregistered, the confirmatory analysis of side choice is conducted on the FN treatment. Of the 149 FN participants, 147 placed at least one bet across the 64 sequences and

¹⁹ This is consistent with the finding from Table 2 that participant heterogeneity strongly dominates sequence-level variation in stake calibration ($\sigma_u/\sigma_s = 0.05$), whereas the opposite holds for betting rates ($\sigma_u/\sigma_s = 3.87$).

are included in the analysis; the remaining two placed no bets and contribute no side-choice observations. Descriptive FN–FP contrasts are also reported ($N = 50$ of 53 FP participants with at least one bet), following the same conventions as for RQ1 and RQ3: posterior medians, 95% credible intervals, and posterior probabilities of direction. Button order (Heads–Tails vs. Tails–Heads) was randomised across trials within each participant. To verify that positional placement did not systematically influence side choice, we conducted a within-participant permutation test ($B = 2,000$ permutations) on the difference in Heads choice rates between the two button-order conditions, restricting to betting trials only. We found no evidence of positional response bias in either treatment (FN: $\Delta = -0.01$, $p = .35$; FP: $\Delta = -0.01$, $p = .69$). Button order was therefore excluded from the primary RQ4 model.

The population baseline Heads probability was $\bar{h} = 0.51$ (95% CI [0.49, 0.53]), indistinguishable from the fair-coin benchmark of 0.50. The distribution of participant-level Heads probabilities was approximately normal, with posterior means averaging 0.51 ($SD = 0.09$) compared with raw participant Heads rates of 0.52 ($SD = 0.13$). The agreement between the raw and posterior means confirms that hierarchical shrinkage regularises individual estimates toward the population baseline without displacing it, as expected under a well-specified model. The narrower posterior spread reflects partial pooling, which is most pronounced for participants with few betting trials. The majority of participants (89.8%) are classified as neutral in terms of individual side bias. Among the remainder, 5.5% are head-biased and 4.7% are tail-biased. (This is a population-level average and does not preclude sequence-level side bias, since a balanced panel would average any such bias out.) The most extreme participants — four classified as solid headish (posterior mean $\hat{\mu}_i^h \in [0.67, 0.88]$) and three as solid tailish ($\hat{\mu}_i^h \in [0.12, 0.15]$) — are consistent side-choosers whose tendencies are well outside the bulk of the distribution. The hierarchical model yields a between-sequence standard deviation of $\hat{\sigma}_s = 0.44$ (95% CI [0.36, 0.55]), indicating that sequences account for a meaningful share of variation in side choice. The participant-to-sequence SD ratio $\hat{\sigma}_u/\hat{\sigma}_s = 1.14$ implies that participant and sequence heterogeneity contribute roughly equally to directional choice probabilities: a markedly different pattern from the economic analysis (Table 2), where individual heterogeneity dominates ($\hat{\sigma}_u/\hat{\sigma}_s = 3.87$ for betting rates and 6.41 for welfare losses).²⁰ This presents an important caveat for the upcoming analysis: while sequences have little economic impact (with the exception of the two pure sequences), they account for meaningful variation in side choice.

Figure 5 plots posterior medians and credible intervals for each sequence, ranked by the posterior probability that their mean Heads choice probability deviates from the population baseline by more than five percentage points ($\delta_{.05}$), and cluster into three roughly equal groups: 20 *Head-biased* (4 strong, 11 moderate, 5 weak), 20 *Tail-biased* (6 strong, 7 moderate, 7 weak), and 24 *neutral*. (The details of this classification are in Online Appendix Table E1.) The top-5 Head-biased sequences are HHTHTT (1.00), HTHHTT (1.00), TTHHTT (0.99), HHHTTT (0.95), and HTHTTT (0.99), and the top-5 Tail-biased are TTHTHH (0.99), THTTHH (1.00), HHTTHH (0.98), TTTHHH (1.00), and THTTTH (0.98), with parenthetical values reporting $\delta_{.05}$. What connects them structurally is that all but one end with a terminal run of at least two of the side opposite to what participants tend to bet on: a pattern that extends beyond the top sequences, holding for 14 of 20 Head-biased and 13 of 20 Tail-biased sequences overall, compared to only 4 of 24 neutral sequences (including, notably, the two pure sequences). The first four pairs are moreover

²⁰ As a descriptive complement, FN–FP contrasts for sequence-level side choices show that the participant-level grand mean difference between treatments is negligible (median = -0.01 , 95% CI [$-0.04, 0.01$]), and the Spearman correlation between FN and FP sequence-level means is moderate ($\rho = 0.54$, 95% CI [$0.37, 0.68$]), indicating that sequences driving directional bias in FN tend to do so in FP as well. The most notable structural difference between treatments is the participant-to-sequence SD ratio: $\hat{\sigma}_u/\hat{\sigma}_s = 0.86$ in FP versus 1.14 in FN—almost exactly opposite margins around parity, with sequence heterogeneity marginally dominating in FP and individual heterogeneity marginally dominating in FN. Most sequences classify as weak or neutral (65%), with only 6 reaching moderate or strong bias, and 49 of 50 participants are classified as neutral—both patterns mirroring FN. The smaller FP sample yields substantially wider credible intervals, limiting the precision of sequence-level estimates, and whether the two sequences with credibly attenuated effects in FP (HHTHTT and TTTHHH) reflect a genuine incentive-frame interaction or sampling variability cannot be determined from these data.

exact mirror images of one another, suggesting that betting against the terminal run reflects a robust response to the underlying sequence pattern rather than idiosyncratic noise. We return to this point in detail in Section 3.4, which draws on both the mirror-image pairing and the terminal-run pattern.

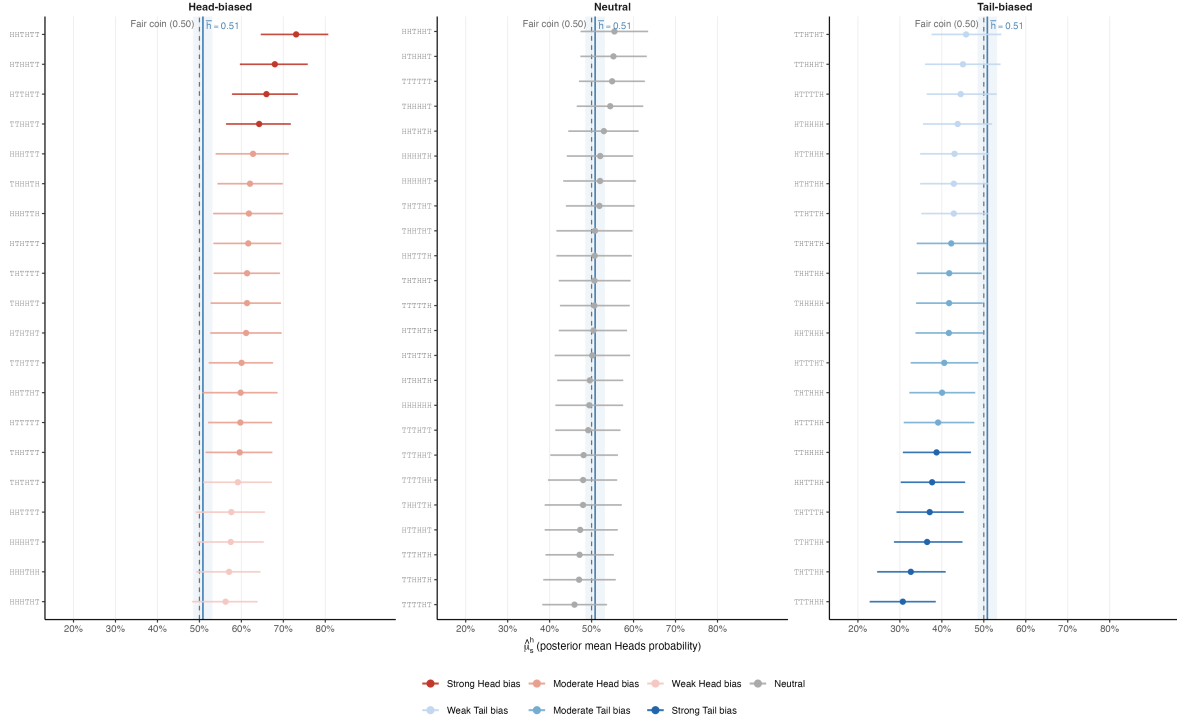


Figure 5: Sequence-level posterior estimates for directional side choice (RQ4, FN treatment), paneled by classification (*Head-biased*, *Neutral*, *Tail-biased*) and ordered within panel by posterior mean Heads-choice probability. Each row corresponds to one of the 64 six-toss sequences. Points show posterior means; horizontal lines show 95% credible intervals. The vertical solid line and shaded band mark the population baseline Heads probability ($\bar{h}$) and its 95% credible interval; the dashed line marks the fair-coin benchmark of 0.50. Colors denote the strength of each sequence’s classification relative to $\bar{h}$ (strong, moderate, weak), separately for Head- and Tail-biased sequences.

Notably, the pure sequences HHHHHH and TTTTTT are themselves classified as neutral, with posterior median Heads-choice probabilities of 49.5% (95% CI [41.4%, 57.5%]) and 54.9% (95% CI [47.0%, 62.7%]), respectively. This finding has direct implications for the preregistered exploratory analysis of the gambler’s and hot-hand fallacies (EX1), which classifies sequences by their distributional similarity to the two pure sequences—serving as behavioral anchors alongside a neutral baseline—on the grounds that the only unambiguous signals of the hot-hand and gambler’s fallacy are, respectively, betting H and T on the H-pure sequence, and T and H on the T-pure sequence. Because the anchors themselves are behaviorally indistinguishable from neutral, the resulting similarity weights carry no information, and the directional index collapses to $\chi \approx 0$ for every sequence: not because participants show no directional tendency, but because the anchors fail to provide a usable reference point. Rather than abandoning the question completely, we classify participants directly from their raw pure-sequence behavior as *gambler-consistent* (bet against the streak on both pure sequences, 37.5%/37.0% FN/FP), *hot-hand consistent* (bet with the streak on both, 31.6%/30.4%), or *inconsistent* (the same side on both, 16.2%/23.9%), and test whether this classification generalizes beyond the two sequences it is built from by exploiting the mirror-pair structure identified in RQ4. Cross-tabulating pure-sequence classification against behavior on the four top-ranked mirror pairs²¹, coding each response as gamblerish if the participant bets against the terminal run and hot-handish otherwise (Table 3), gambler-consistent participants prove strikingly coherent: 88.2% (70.6% in FP) also respond gamblerish

²¹ HHTHTT–THTTHH, HTHHTT–THTTHH, THTHTT–HHTTHH, and HHHTTT–TTTHHH

on the mirror sequences. Hot-hand consistent participants, by contrast, are nearly evenly split: 52.4% gamblerish and 38.1% hot-handish on mirrors (46.2% and 38.5% in FP). The raw Heads-choice probabilities on the pure sequences clarify the nature of the inconsistent group: these participants bet Heads on both H-pure and T-pure at approximately 59%, consistent with an unconditional side preference rather than streak-based reasoning. This asymmetry suggests gamblerishness is fairly robust, consistent with low individual heterogeneity in whatever mechanism produces it, whereas the near-even split among hothandish participants looks closer to noise, obscuring rather than reflecting a single coherent disposition. Whether this reflects two genuinely distinct psychological processes, or a single process that expresses itself differently depending on which cue a sequence happens to provide, is a question we return to below in Section 4. Full documentation of the EX1 collapse, its geometric mechanism, and the pure-sequence classification’s covariate associations is in Online Appendix C.2.

Pure-sequence type	n	%	Mirror-pair behavior (%)			$\bar{p}_H$ on pure seq.	
			Gamblerish	Hot-handish	Split	HHHHHH	TTTTTT
Hot-hand consistent	43	31.6	52.4	38.1	9.5	1.00	0.00
Gambler consistent	51	37.5	88.2	5.9	5.9	0.00	1.00
Inconsistent	22	16.2	47.6	23.8	28.6	0.59	0.59
Unclassified	20	14.7	—	—	—	0.56	0.55
Total	136	100.0					

Table 3: Pure-sequence classification and mirror-pair consistency (FN treatment). Hot-hand consistent: streak continuation on both pure sequences; gambler consistent: streak reversal on both; inconsistent: same side on both; unclassified: missing bet on at least one pure sequence. Mirror-pair behavior is coded as gamblerish if the participant bets against the terminal run on the four top-ranked mirror pairs from RQ4. $\bar{p}_H$: raw proportion of Heads bets on the indicated pure sequence. Mirror-pair percentages exclude the unclassified group. FP results: gambler consistent 70.6% gamblerish; hot-hand consistent 46.2% gamblerish, 38.5% hot-handish.

Another part of the analysis affected by the GHI collapse was the preregistered comparison between anchor-based directional classifications and canonical rule-based heuristics from the literature, which cannot be executed as intended since $\chi_s \approx 0$ for all sequences. Correlating the canonical sequence-level scores $Q_s^{(k)}$ directly against the RQ4 estimates μ_s^h instead (Online Appendix Table C1), all Spearman correlations are near zero and statistically indistinguishable from chance in both treatments—not a power issue, but a structural one: because participants bet against the terminal run on virtually every sequence where one is present, the run-length rule is uniformly negative wherever it applies, leaving no variance left to correlate with μ_s^h . The rule is nonetheless a useful coarse heuristic: no sequence in either treatment credibly elicits the opposite, hot-handish tendency, so where the rule fires it is more likely than not correct, and its worst-case error is a false positive rather than a sign reversal. See more details in Online Appendix C.3.

The last preregistered analysis concerns whether two individual-difference measures, dispositional optimism and response time, are associated with participants’ betting rate ($\hat{\mu}_Z^b$), stake deviation ($\hat{\mu}_Z^a$), welfare loss ($\hat{\mu}_Z^c$), and baseline Heads-choice probability ($\hat{\mu}_Z^h$). Full posterior estimates for both predictors, including between-outcome heterogeneity and residual variation in each treatment, are reported in Online Appendix Table D4. Dispositional optimism shows no credible association with any behavioral outcome in either treatment. Across all four outcomes the credible intervals for the optimism slopes straddle zero and the pooled estimates are small and imprecise ($\bar{\beta}^{\text{opt}}$: FN = 0.05 [−0.10, 0.18], FP = −0.06 [−0.29, 0.20]), with negligible between-outcome heterogeneity (τ^{opt} : FN = 0.05, FP = 0.12). Response time is a more consistent predictor. Slower participants rank lower on all three economic margins in both treatments — lower betting probabilities ($\hat{\mu}_Z^b$), larger stake deviations ($\hat{\mu}_Z^a$), and higher welfare losses ($\hat{\mu}_Z^c$) — with a credibly negative pooled slope in FP ($\bar{\beta}^{\text{rt}}$ = −0.30 [−0.50, −0.09], $P(>0) = 0.01$) and a directionally consistent but imprecise estimate in FN (−0.15 [−0.34, 0.06], $P(>0) = 0.05$). The

heads-bias outcome ($\hat{\mu}_Z^h$) is flat in FN ($-0.05 [-0.19, 0.12]$, $P(>0) = 0.27$) but credibly negative in FP ($-0.29 [-0.46, -0.10]$, $P(>0) = 0.00$), the only outcome where the two treatments diverge qualitatively.

3.4 Synthesis

The preceding analyses have discovered that certain sequences are economically costly, especially the two pure ones, and that certain sequences systematically shift participants' directional choices, especially those ending in a terminal run. Whether the costly sequences are also the ones that psychologically bias side choice, and whether the terminal run is the only cue that elicits such bias, are the questions we now turn to. Unlike Sections 3.2 and 3.3, the analyses in this section were not preregistered: sequences are selected post-hoc based on their posterior estimates from those two sections, so the patterns reported below should be read as descriptive and hypothesis-generating rather than as independent confirmatory tests. To begin with, we compute, for the five sequences with the largest welfare losses and the five most head-biased and five most tail-biased sequences, the deviation of each previously defined sequence-level quantity's posterior median from its corresponding grand mean: $\Delta_s^h = \hat{\mu}_s^h - \bar{\mu}^h$, $\Delta_s^c = \hat{\mu}_s^c - \bar{\mu}^c$, $\Delta_s^b = \hat{\mu}_s^b - \bar{\mu}^b$, and $\Delta_s^a = \hat{\mu}_s^a - \bar{\mu}^a$, for heads-choosing, welfare loss, betting, and staking, respectively. Table 4 reports these percentage-point deltas, enabling a direct comparison of the sequences' relative effects across all four dimensions on a common scale.

	<i>Heads-share</i> Δ $\bar{\mu}^h = .51$	<i>Welfare-loss</i> Δ $\bar{\mu}^c = .12$	<i>Betting-rate</i> Δ $\bar{\mu}^b = .74$	<i>Stake-deviation</i> Δ $\bar{\mu}^a = .07$
<i>Top-5 welfare-losing sequences</i>				
HHHHHH	-.01	+.17*	+.12*	+.12*
TTTTTT	+.04	+.13*	+.12*	+.09*
HHTTHT	+.09*	+.04*	-.19*	+.00
HTHTHT	+.10*	+.03*	-.03	+.01
HHTHHH	-.09*	+.01	+.03	+.03
<i>Top-5 head-biased sequences</i>				
HHTHTT	+.22*	+.00	-.08*	-.02
HTHHTT	+.17*	+.00	-.09*	-.01
HTTHTT	+.15*	+.00	+.07*	+.00
TTHHTT	+.13*	-.02*	+.07*	-.02
HHHTTT	+.12*	-.01	-.04	+.00
<i>Top-5 tail-biased sequences</i>				
TTTHHH	-.20*	+.00	-.09*	+.00
THTTHH	-.18*	-.01	-.11*	+.01
TTHTHH	-.14*	-.01	-.11*	-.01
THTTTH	-.14*	-.01	+.01	+.00
HHTTHH	-.13*	-.02*	+.07*	+.00

Table 4: Each cell reports the percentage-point delta between the posterior median for the sequence shown in the row (computed over all participants) and the corresponding grand mean for the quantity shown in the column. Grand means are computed across all participants and sequences and are reported below the column labels. All underlying quantities are percentages: *Heads share* and *Betting rate* are measured on the probability scale, whereas *Welfare loss* and *Stake deviation* are expressed as shares of the endowment. All quantities are estimated using data from the FN treatment. *Heads share* uses the full sample; the other three quantities use the confirmatory sample. An asterisk denotes a 95% credible interval excluding zero.

Table 4 reveals a surprisingly clear pattern: meaningfully side-biased sequences are associated with little to no relative welfare loss, whereas the two (pure) sequences with the largest welfare-loss deltas show no side bias. Further inspection of the table provides two additional insights. First, the 12-percentage-point increase in the betting-rate delta for the pure sequences moves their betting rates closer to the normative optimum of universal betting, making them actually *the least* biased in terms of betting rates. Their welfare losses are therefore attributable primarily to stake deviations, corroborating findings from

Section 3.2. Second, the direction of the betting-rate deviation sometimes aligns with the direction of side bias and sometimes runs counter to it. For example, six of the ten most side-biased sequences have betting-rate deltas of up to 11 percentage points below the grand mean, whereas the remaining four have deltas of up to 7 percentage points above it. Importantly, this mix does not look random. Revisiting the mirror-pair pattern identified in Section 3.3 reveals that, within each pair, the two sequences share the same direction of the betting-rate shift. However, some pairs shift betting rates upward, whereas others shift them downward. Consider the pairs TTHHTT–HHTTHH and HTHHTT–THTTHH. Exposure to the former pair increases the betting rate, whereas exposure to the latter decreases it. These observations raise the possibility that side choice and the decision to bet are dissociated: each may be driven by a distinct feature of the sequence rather than by a single underlying signal.²² To explore this dissociation, Figure 6 plots the betting-rate delta against the side-bias delta for all 64 sequences, connecting the two members of each mirror pair.

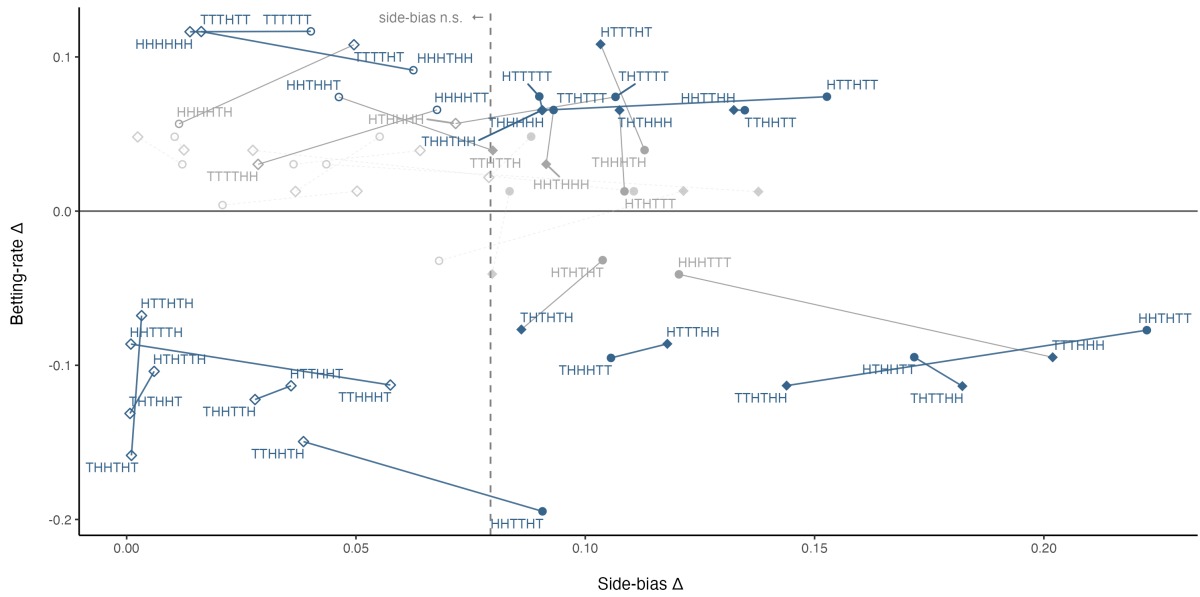


Figure 6: The figure plots sequences’ *Betting-rate* Δ against the *Side-bias* Δ (defined as the absolute value of *Heads share* Δ). The underlying data are the same as in Table 4, which reports only a subset of 15 of the 64 sequences shown here. Sequences are shown in blue and gray color: blue indicates the sequence’s betting rate delta is credibly different from zero, gray indicates the opposite. Mirror pairs are connected by solid and dashed line segments: blue solid line indicates that both of the pair are in blue, and gray solid line indicates that only one of the mirror pair is in blue—in this case, both sequences have both label and symbol. Dashed gray line indicates that both of the mirror pair is in gray, and in that case, both sequences only have a symbol, but not label. The symbols indicate which side participants predominantly bet on: heads are denoted by circles ($\circ/\bullet$) and tails by diamonds ($\diamond/\blacklozenge$). Fill of the symbols indicates whether the side-bias is credibly different from a random choice (filled symbols, $\bullet/\blacklozenge$) or it is indistinguishable from a random choice (empty symbols, $\circ/\diamond$). The vertical dashed gray line is an approximate boundary of the side-bias significance—sequences on the left from it do not elicit a side-bias and the ones on the right do. A sequence’s side bias is classified as significant when the 95% credible interval of its posterior difference from the grand mean excludes zero.

These are the patterns Figure 6 reveals. (i) Betting rate rises with the imbalance, with posterior-average rates climbing from 63.7% for balanced sequences (all of them exhibiting below-average betting,

²² An alternative, mechanical explanation is selection: because side choice is observed only among participants who bet, sequences with lower betting rates may leave a more selective group of bettors. If those who choose to bet are more “certain” and therefore more side-biased, lower betting rates could mechanically be associated with stronger observed side bias. The data do not support this explanation at the sequence level. Because imbalance raises betting while lowering side bias, the unconditional correlation ($\rho = -0.003$, $p = .98$) is uninformative; conditional on imbalance class and the terminal run, however, sequences with lower betting rates still do not display stronger side bias ($\beta = 0.22$, 95% CI $[-0.10, 0.53]$; no within-stratum correlation is significant). At the participant level, less eager bettors do respond more strongly to imbalance in their side choice ($\rho = -0.35$, 95% CrI $[-0.43, -0.26]$), but because these responses split between following and countering the majority, they do not add up to sequence-level side bias; and for the terminal run, which drives most side bias, betting propensity and directional response are unrelated ($\rho = 0.01$, $[-0.22, 0.23]$). Full test statistics, including a formal bimodality test, are reported in Online Appendix C.8.

18 significantly so) to 77.5% for 2–4 sequences, 80.8% for 1–5, and 85.6% for 0–6. Note that the jump is not linear: the largest increase is between balanced and imbalanced, and the steps between the various degrees of imbalance are modest. (ii) Side bias is predominantly connected with sequences ending in a terminal run of at least two identical symbols:²³ of the 32 sequences ending in a terminal run of at least two identical symbols, 23 produce a significant side bias, averaging 12.3 percentage points, while the remaining 9 do not, averaging 5.0pp. The direction of the bias is against the run in 30 out of 32 run-ending sequences, including the 23 significantly side-biased ones. (iii) Side bias is also negatively associated with imbalance: among run-ending sequences, the average side bias is 9.6 percentage points for the least imbalanced sequences (2–4), 7.8pp for 1–5, and just 2.7pp for the pure sequences. Regarding direction, 15 of the 20 significantly side-biased imbalanced sequences elicit bets against the majority symbol, compared with 22 of all 44 imbalanced sequences. (iv) The side bias is strongest, however, in the eight balanced sequences with a terminal run, averaging 15.9pp. Of the 12 significantly side-biased balanced sequences, these eight are exactly the ones with a terminal run, and all eight elicit bets against the run. (v) Consequently, among significantly side-biased sequences, balanced ones elicit significantly greater side bias than imbalanced ones: 14.1 vs. 10.6 percentage points on average ($p = .019$, one-sided Welch’s t -test; $p = .025$, Mann-Whitney). (vi) Alternation rate shows no association with betting rate or side bias.²⁴ (vii) In the majority of cases (24 of 32), the two members of a mirror pair fall in the same quadrant, indicating that betting and side-choice behavior is governed by the sequence’s abstract structure rather than by the specific symbol composition.²⁵

The common apparent theme underlying these patterns is, at the aggregate level, *balance restoration*—75% of the significantly side-biased imbalanced sequences exhibit a bias toward restoring balance. (This aggregate pattern should not be read as a direct response to imbalance, however. Among imbalanced sequences, a terminal run almost always consists of the majority symbol (22 of the 24 run-ending imbalanced sequences), so betting against the run and betting to restore balance largely coincide; the joint model below separates the two cues and shows that, holding the run fixed, imbalance on average pulls choices *toward* the majority.) Sequences that elicit the strongest balance-restoration response are located in the upper-right corner, predominantly sharing two properties—they are the least imbalanced (that is, 2–4) and always end in a run of two identical symbols against which participants bet. Moving westward from this region, side bias decreases and the betting rate increases, both as a function of sequence imbalance. This indicates that increasingly imbalanced sequences attract more participants to bet, yet the aggregate side bias diminishes. Thus, the additional bettors and/or a subset of the original bettors must be shifting away from balance restoration and toward imbalance amplification. We cannot determine from our data the precise mechanism underlying this

²³ Only the presence of a terminal run matters, not its length: holding imbalance fixed, runs of two and runs of three or more produce practically identical side bias (9.6 vs. 9.6pp among 2–4 sequences; 15.8 vs. 16.1pp among balanced sequences). The pure sequences, whose run spans all six symbols, are the only exception, but there run length cannot be separated from maximal imbalance.

²⁴ Two pairs are noteworthy, still. THTHTH–HTHTHT are the fully alternating sequences, with five switches, and they are the only balanced no-run pair to elicit a side bias. HHHHHH–TTTTTT, on the other hand, are the no-alternation sequences, with zero switches, and they are among a handful of imbalanced sequences with a terminal run that do not produce a side bias. Tellingly, the former look “too random” and the latter not random at all. Interestingly, since alternation rate is correlated with imbalance ($\rho = -0.48$, $p < .001$), not correcting for this confound may deceive one into finding a spurious effect of alternation on betting rate. More generally, the 12 balanced sequences without a run have the highest alternation rates in the design, and the few among them that tilt side choice do so toward extending their own pattern: THTHTH and HTHTHT toward continuing the alternation, HHTTHT and TTHHTH toward extending their pairing. Pattern-seeking thus does not disappear when neither cue is present; it shifts to whatever structure remains. Response times offer no direct support for this interpretation, however. Sequences with more alternations do take longer to decide on (FN: Spearman $\rho = 0.43$, $p < .001$; FP: $\rho = 0.17$, $p = .18$), but the 12 balanced sequences without a run are not slower as a group (FN: +1.0%, 95% CI [–2.5%, 4.8%]), and taking longer than usual on a trial does not make a pattern-completing bet more likely (+1.0pp per 50% longer response time, $p = .42$). Longer decision times on highly alternating sequences thus more plausibly reflect the effort of reading them than a search for structure.

²⁵ No pair shows a significant divergence in betting direction, and none of the 8 exceptions involve one member overbetting and the other underbetting significantly; most instead cross the side-bias significance threshold, and several of these crossings are themselves borderline—the losing member’s credible interval sits within 0.015 of zero in 6 of the 8 cases.

shift, but some participants appear to act *as if* highly imbalanced sequences indicated a malfunctioning random generator—an interpretation echoed by the minority of participants who describe betting with the more frequent symbol in their debrief (see below). Moving south-eastward from the region of strongest balance restoration, we encounter sequences that are balanced and end in a terminal run. These sequences elicit the strongest side bias despite having below-average betting rates. This presents a puzzle—if some participants simply found the balanced sequences less attractive to bet on, we would expect a lower betting rate but a similar, rather than higher, side bias. Because selection does not account for this pattern (footnote above), the observed behavior points to two distinct cues acting at once: the balanced composition discourages some participants from betting, while the terminal run gives those who do bet a clear direction. To explain this behavior, we propose that betting against the terminal run not only restores balance locally, but also globally, through a simple rule of thumb: terminal run signals imbalance. This is a universal property of sequences of any length.²⁶ For six-symbol sequences specifically, 75% of sequences with terminal run are also imbalanced. Such a rule would be a prime example of Simon’s bounded rationality and satisficing, as it would allow individuals to assess a sequence’s balance or imbalance with relatively high precision without having to analyze anything beyond its two final symbols. There is also a strong ecological rationale for such a rule: in nature, analogous series are typically observed sequentially rather than as complete histories. We may thus have at least two groups of participants: naive participants, who assess imbalance using the simple run rule, and sophisticated participants, who either corroborate the run rule against the overall imbalance or do not use the rule at all. Finally, in the last undescribed segment, in the lower-left corner, we find balanced sequences with no terminal run. These sequences elicit below-average betting rates and no side bias. This should no longer be surprising: they provide participants with no salient cue from which to infer a direction, leaving them either to refrain from betting or to choose a side at random. What we have described at length in verbal terms can be formalized as the result of two competing forces: the run rule, which universally prescribes “bet against the terminal run”, and the assessment of overall imbalance, which, depending on participants’ traits, prescribes either “bet to restore balance” or “if the imbalance is high, bet to maintain the imbalance”. When these rules coincide, the result is a high betting rate and either high or low side bias, depending on the composition of participant types. When only one of the two signals is present, the result is low engagement but high side bias; when both signals are absent, both betting engagement and side bias are low.

In order to assess the relationship between these two rules, we run a hierarchical model that jointly estimates both whether to bet and what side to choose. The betting equation regresses the decision to bet on imbalance magnitude and a terminal-run dummy; the side-choice equation regresses side choice, conditional on betting, on signed imbalance and the signed terminal run. Both equations include participant-specific random intercepts and slopes on their respective regressors, together with sequence-level random intercepts, but rather than being estimated separately, the two equations’ participant-level random effects—intercept, imbalance-slope, and run-slope, in each equation—are drawn jointly from a single multivariate-normal distribution with a freely estimated correlation matrix. The random-effect variances recover each margin’s individual heterogeneity, while the correlation matrix’s cross-equation entries—between a participant’s betting intercept and side-choice intercept, and between their engagement- and direction-slopes on each cue—test whether engagement and

²⁶ Let $N = 2k$ and split a sequence into its last two symbols and the preceding $(N - 2)$ -symbol prefix. If the last two symbols differ, the full sequence is balanced if and only if the prefix is balanced; if they are identical (a terminal run), it is balanced if and only if the prefix is exactly one step away from balance. Under a fair coin, $P(\text{balanced}) = \binom{2k}{k}/2^{2k}$ and $P(\text{balanced} \mid \text{run}) = \binom{2k-2}{k-2}/2^{2k-2}$, and their ratio is $2(k-1)/(2k-1) < 1$. A terminal run therefore always lowers the probability of balance and, since $P(\text{balanced}) \leq 1/2$ for every even N , a run-ending sequence is always more likely imbalanced than balanced. The ratio rises toward, but never reaches, 1 as $N \rightarrow \infty$, so the cue is most informative for short sequences; for $N = 6$, $P(\text{balanced} \mid \text{run}) = 1/4$ against $5/16$ unconditionally.

direction reflect one underlying disposition or genuinely separate processes. Formal description of the model, together with the implementation details, is in Online Appendix C.4. Table 5 supports the dissociation account with respect to direction: neither participant-level slope correlates credibly across equations (ρ panel), so how strongly a cue draws a participant into betting is unrelated to which side it pushes them toward. The dissociation is not complete, however: participants whose betting responds more strongly to imbalance also respond more strongly to it in their side choice, in whichever direction they lean (correlation with the magnitude of the side-choice slope: $\rho = 0.47$, 95% CrI [0.36, 0.58]; FP: 0.46, [0.26, 0.62]), whereas no such link exists for the terminal run ($\rho = 0.29$, [-0.17, 0.63]).²⁷ The decision to bet is driven almost entirely by the imbalance magnitude, on average increasing the probability of betting by 21.1pp across its full range (FP: 24.7pp); the core pattern is broadly consistent across the multiplier conditions, with the remaining differences plausibly reflecting individual heterogeneity, to the analysis of which we now turn. For side choice, the two cues instead pull in opposite directions: moving from a sequence without a terminal run to one ending in a run shifts the probability of choosing heads by 13.6pp toward the side opposite the run, while moving from a balanced sequence to one fully imbalanced in one direction shifts the probability toward the majority side by 8.0pp. (The two forces are structurally unequal. The run’s pull is fixed—it applies in full whenever a run is present—whereas imbalance’s pull scales with the degree of imbalance and reaches its maximum only at the pure sequences. For every run-ending sequence, the run therefore dominates, which is why betting against the run prevails even on the most imbalanced sequences and why the aggregate pattern looks like balance restoration. In terms of the familiar labels, side choice follows the hot hand (backing the majority) only when no run is present, and switches to the gambler’s fallacy (betting against the streak) as soon as one appears.)

Online Appendix Table C3 quantifies this individual heterogeneity, revealing three patterns. In side choice, the terminal-run slope is comparatively homogeneous ($\tau = 1.00$ in FN, 1.06 in FP; $\tau/|\hat{\beta}| = 1.07$ and 1.34), so the population fixed effect is a reasonably representative summary of individual behavior; in the betting equation, the run slope varies little across participants in absolute terms ($\tau = 0.30$ and 0.45), around a fixed effect close to zero. The imbalance slope in the betting equation shows the same qualitative pattern despite substantially larger absolute heterogeneity ($\tau = 5.10$ in FN, 4.08 in FP): because its fixed effect is proportionally large enough, the ratio stays modest ($\tau/|\hat{\beta}| = 1.69$ and 1.86), so this cue is not expected to split bettors into distinct types either. The most striking heterogeneity is the imbalance slope in the side-choice equation: its SD is comparable in magnitude to the betting equation’s ($\tau = 4.76$ in FN, 4.68 in FP), but because the corresponding fixed effect is far smaller, the same absolute heterogeneity explains far more relative to it ($\tau/|\hat{\beta}| = 6.40$ in FN, 13.59 in FP) – consistent with participants differing in the *sign* of their response to imbalance, not only its strength.

To illustrate the heterogeneity structure for side choice, Figure 7 plots posterior medians of participants’ personal run- and imbalance-slopes for both treatments. The left panel shows that side-choice run-slopes are concentrated to the left of zero in both treatments, suggesting most participants choose against the terminal run, and essentially none choose with it. The right panel

²⁷ Side-choice slopes are signed (toward Heads, or toward the majority symbol), whereas betting slopes are unsigned. For the run cue this is immaterial, because nearly all participants bet against the run; the signed cross-equation correlation is not credible in either treatment (FN: $\rho = -0.33$, [-0.72, 0.21]; FP: $\rho = -0.27$, [-0.71, 0.28]). For imbalance, where participants split into majority-followers and balance-restorers, the signed correlation is null (FN: $\rho = -0.06$, [-0.26, 0.14]; FP: $\rho = -0.17$, [-0.46, 0.14]), but the correlation with the slope’s magnitude reported in the text is not. Magnitude correlations are computed from participants’ personal effects within each posterior draw; because side-choice slopes are better identified for participants who bet mainly on imbalanced sequences, part of this association may reflect estimation precision rather than disposition. The only credible cross-equation intercept correlation is in FP ($\rho = 0.45$, [0.02, 0.76]; FN: $\rho = -0.17$, [-0.40, 0.07]): more eager bettors also lean toward Heads, a baseline label preference unrelated to the sequence cues. Within equations, participants who bet more overall respond less to imbalance when deciding whether to bet (FN: $\rho = -0.31$, [-0.51, -0.08]; FP: $\rho = -0.28$, [-0.57, 0.07]), and participants who bet more strongly against the run also tend to bet against the majority (FP: $\rho = 0.39$, [0.04, 0.66]; FN: $\rho = 0.19$, $P(\rho > 0) = .94$), in line with Table 6.

	FN Treatment			FP Treatment		
	Slope	AME (pp)	Cohen's h	Slope	AME (pp)	Cohen's h
<i>Betting equation (Bet = 1)</i>						
Intercept	1.67 [1.15, 2.24]	67.5% [66.0, 69.0]	—	0.68 [−0.18, 1.63]	57.5% [55.5, 59.6]	—
Terminal run {= 1}	0.10 [−0.10, 0.31]	+0.6 [−1.2, 2.2]	0.02 [−0.03, 0.05]	0.26 [−0.02, 0.55]	+2.8 [0.2, 5.0]	0.06 [0.00, 0.11]
Imbalance magnitude {0, $\frac{1}{3}$, $\frac{2}{3}$, 1}	3.01 [2.01, 4.05]	+21.1 [17.7, 24.3]	0.36 [0.29, 0.44]	2.19 [0.99, 3.42]	+24.7 [20.1, 29.3]	0.38 [0.28, 0.47]
<i>Side-choice equation (Heads = 1)</i>						
Intercept	0.08 [−0.06, 0.23]	51.9% [50.1, 53.6]	—	0.03 [−0.19, 0.24]	50.7% [47.6, 53.8]	—
Terminal run {−1 = TT , 0, 1 = HH }	−0.93 [−1.18, −0.69]	−13.6 [−16.1, −11.2]	−0.27 [−0.32, −0.22]	−0.79 [−1.20, −0.38]	−12.0 [−15.5, −8.2]	−0.24 [−0.31, −0.16]
Imbalance {−1 (T) to 1 (H), by $\frac{1}{3}$ }	0.74 [−0.07, 1.56]	+8.0 [3.3, 12.0]	0.16 [0.08, 0.23]	0.34 [−0.96, 1.65]	+7.6 [0.4, 13.6]	0.13 [0.01, 0.23]
<i>Participant-level correlation between equations (ρ, Bet $\leftrightarrow$ Heads)</i>						
Intercept		−.17 [−.40, .07]			.45 [.02, .76]	
Terminal run		−.33 [−.72, .21]			−.27 [−.71, .28]	
Imbalance		−.06 [−.26, .14]			−.17 [−.46, .14]	

Table 5: **Population-average effects of the terminal run and imbalance, and cross-equation dissociation test, by treatment.** Each cell reports the posterior median with its 95% credible interval in brackets below. Curly braces {} beneath each regressor give its coding or admissible value set. FN: 149 participants; FP: 53 participants. *Slope* is the log-odds coefficient of the fixed effect. *AME* is the population-averaged marginal effect on the probability scale, integrating over the fitted participant- and sequence-level random effects rather than assuming them at zero. For the Intercept rows it is the RE-integrated baseline predicted probability (no terminal run, zero imbalance). For the other rows it is the instantaneous per-unit marginal effect (dp/dx) evaluated at each observation's actual data: because the betting-equation regressors are coded 0 to 1, "per unit" spans their full observed range; because the side-choice regressors are coded −1 to 1, "per unit" corresponds to moving from neutral (no cue) to the cue fully present in one direction, half of the total coded range. *Cohen's h*, $2 \arcsin \sqrt{p_1} - 2 \arcsin \sqrt{p_0}$, is computed per posterior draw from the discrete change in probability when the cue moves from its reference to its treated level on the actual trials (run present vs. absent, and fully imbalanced vs. balanced, for betting; the probability of choosing the run's symbol and the majority symbol, respectively, for side choice). For imbalance and betting this discrete change is 15.4pp (FN) and 17.5pp (FP), below the AME, which is a per-unit slope. AME and h are evaluated at the estimated random effects of the participants and sequences in the sample, so their intervals need not coincide with those of the slope. The ρ panel reports the participant-level correlation between the corresponding random slopes or intercepts in the two equations.

shows that imbalance-slopes have most of their mass near zero but fat tails in both directions, with, if anything, slightly more mass on the positive (majority-following) side than the negative (minority-following) side. This pattern suggests a near-uniform response to the run cue, but a response to imbalance whose sign differs across participants—some follow it, promoting the imbalance, and others counter it, reducing it. Panel B of Online Appendix Table C3 classifies participants with a credible personal slope on the run and imbalance cues into three categories based on these slopes – *Against* (imbalance/run), *With* (imbalance/run), and *Uncertain* – corresponding to a slope that is credibly negative, credibly positive, or indistinguishable from zero at the 95% level, respectively, estimated separately by treatment. For the run cue, 40% of participants choose against the cue, only 1% with the cue, and 59% are classified as uncertain (40%, 2%, and 58% in the FP treatment). For the imbalance cue, 19% choose against the imbalance, 30% with the imbalance, and 51% are uncertain (19%, 28%, and 53% in the FP treatment).

An obvious question now arises: do the two classifications overlap? The answer speaks directly to the proxy-mechanism account proposed earlier in this section: if betting against the terminal run functions, for at least some participants, as a rule-of-thumb stand-in for assessing overall imbalance, then

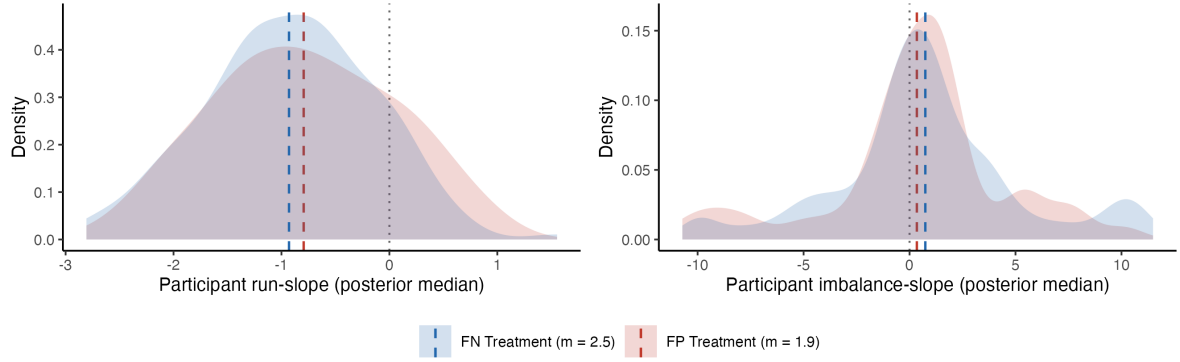


Figure 7: **Individual heterogeneity in side-choice run- and imbalance-slopes, by treatment.** Each panel shows the density of participants’ personal slopes (population fixed effect plus their own random-effect draw, posterior median), separately for the FN (blue, $m = 2.5$) and FP (red, $m = 1.9$) treatments. The grey dotted line marks zero; the colored dashed lines mark each treatment’s population fixed effect. Left: the run-slope, where negative values indicate choosing against the terminal-run side and positive values choosing with it. Right: the imbalance-slope, where positive values indicate choosing the majority side (extending the imbalance) and negative values the minority side (restoring balance).

participants classified as *Against Imbalance* should rely on the run cue more heavily than those classified as *With Imbalance*. Splitting participants by this classification (Table 6), *Against Imbalance* participants bet against the terminal run roughly twice as strongly as *With Imbalance* participants (median run-slope -1.53 vs. -0.74 ; Mann-Whitney $p < .001$) and more strongly than the *Uncertain* group (-0.82 ; $p < .001$), whereas *With Imbalance* participants are statistically indistinguishable from *Uncertain* on this dimension ($p = .59$)—absent an imbalance-restoring disposition, the run heuristic is applied no more forcefully than by participants with no discernible imbalance preference at all. A rank correlation between the continuous personal run- and imbalance-slopes across all 149 FN participants points the same way (Spearman $\rho = 0.22$, $p = .007$), consistent with the model’s own within-equation correlation ($\Omega_{\text{run,imb}} = 0.19$, 95% CrI $[-0.05, 0.40]$, $P(\Omega > 0) = .94$). Additionally, the run-slope also covaries with response time: those who take longer to decide lean more strongly against the terminal run and a magnitude-based check indicates this reflects genuinely stronger run-cue engagement among slower deliberators rather than general disengagement. The imbalance-slope’s magnitude, by contrast, is credibly and negatively associated with both dispositional optimism and risk preference: more optimistic and more risk-averse participants rely less on the imbalance cue overall, regardless of which way they lean when they do respond to it.²⁸ Together, these results suggest the run rule is not applied uniformly across participants irrespective of their broader relationship to imbalance — it is applied most forcefully by exactly the participants whose imbalance behavior it is theorized to proxy for, and weakly by those it should not predict at all. The association is necessarily correlational; thus, it cannot establish that participants literally derive the run rule from an imbalance assessment, rather than the two behaviors sharing some common underlying sensitivity to perceived non-randomness.

The run-slope comparison above corroborates the imbalance classification from within the model itself — it behaves the way the proxy account predicts. A second, independent check is whether the classification means anything to participants themselves: we compare the three model-derived slope classes to participants’ self-reported strategy use from the post-task questionnaire (Online Appendix F.1.10), which asked whether participants had followed a rule or

²⁸ Full results in Online Appendix Table D5. Signed run-slope $\times$ response time: $\rho = -0.29$ $[-0.38, -0.18]$ in FN, -0.29 $[-0.45, -0.13]$ in FP (both $P(\rho < 0) > 0.99$). Magnitude check: $|\text{run-slope}| \times \text{response time}$ is credibly positive in both treatments ($\rho = 0.25$ $[0.14, 0.35]$ in FN, 0.25 $[0.05, 0.43]$ in FP), while $|\text{imbalance-slope}| \times \text{response time}$ is not, consistent with run-specific engagement rather than general attentiveness—though the run and imbalance cues are themselves moderately correlated across the sequence set ($r = 0.58$, $n = 64$ sequences), so this asymmetry is suggestive rather than conclusive. Imbalance-slope magnitude $\times$ optimism: $\rho = -0.15$ $[-0.23, -0.07]$ in FN, -0.23 $[-0.36, -0.09]$ in FP. Imbalance-slope magnitude $\times$ risk preference: -0.11 $[-0.19, -0.03]$ in FN, -0.19 $[-0.34, -0.04]$ in FP.

Table 6: Run-slope by imbalance-slope classification (m25, $n = 149$)

Imbalance class	n	Median run-slope	% credibly against-run
Against Imbalance	28	1.52	60.7%
Uncertain	76	0.83	36.8%
With Imbalance	45	0.74	33.3%

Note: Against Imb. vs. With Imb., Mann-Whitney $p < .0001$; Against Imb. vs. Uncertain, $p < .001$; With Imb. vs. Uncertain, $p = .60$ (n.s.). Spearman correlation between the two continuous slopes across all 149 participants: $\rho = 0.22$, $p = .007$.

strategy when making decisions (Question 8) and, if so, to describe it in their own words (Question 9). The main finding is that having answered Question 8 (yes, no, or unsure) is not predictive of the slope-based classes for either the imbalance or the run cue (Online Appendix Table C7). However, restricting attention to the 97 participants who reported having used a strategy (“Yes” in Question 8) and coding their free-text description of it (Question 9) into five categories (see the coding procedure in Online Appendix C.5) yields a tight correspondence for the imbalance cue. As Table 7 shows, participants who described betting against the more frequent symbol are almost always classified as *Against Imbalance* (14 of 17), and all 12 participants who described risk-based or independence-based reasoning, with no directional pattern, are classified as *Uncertain*. Participants who described betting with the more frequent symbol are also predominantly classified as *With Imbalance* (3 of 4), though this category rests on very few observations and should be read with that caveat. For the run cue, the analogous comparison is not informative: almost no participant’s fitted slope falls in the *With Run* direction (1 of 97 in the pooled sample), so there is no meaningful with/against split left to validate against the free-text descriptions (see Online Appendix Table C8) — consistent with the near-universal population-level tendency to bet against the terminal run documented above, rather than a weakness of the debrief coding itself. This correspondence, most robust for the anti-trend and no-trend categories, is reassuring: it suggests that the imbalance-slope classification captures a genuine, self-recognized feature of participants’ strategies rather than an artifact of the estimation procedure, even though the run-cue comparison above cannot speak to this either way.

Category	With Imbalance	Uncertain	Against Imbalance	OR (95% CI)	p
Pro-trend	3	1	0	11.29 (0.85, 618.82)	0.035
Anti-trend	0	3	14	25.08 (5.86, 156.70)	< 0.001
No-trend	0	12	0	∞ (3.29, ∞) [†]	< 0.001
Misc	4	16	5	—	—
Unclear	15	17	7	—	—
Total	22	49	26		

Table 7: Self-reported strategy category versus model-derived imbalance-slope class (pooled across treatments, $n = 97$). Odds ratios, 95% CIs, and p -values are two-sided Fisher exact tests, each collapsing the row to category-vs-rest and the predicted column to that class-vs-rest; reported only for the three categories with an a priori directional prediction.

[†]All 12 no-trend participants fall in *uncertain*; the 2×2 table has a structural zero, so the point estimate of the odds ratio is infinite (only the lower confidence bound is informative). Misc and Unclear have no a priori predicted class and are reported descriptively only. Omnibus (3×3) Fisher exact test for the full table: $p < 0.001$.

The run and imbalance cues analyzed above imply distinct, testable predictions. Heterogeneity in the imbalance effect is concentrated in side choice rather than in the decision to bet ($\tau/|\hat{\beta}| \approx 1.7$ –1.9 for betting vs. 6.40–13.59 for side choice), so the imbalance cue should be a strong predictor of whether participants bet, but not of which side they choose, absent knowledge of individual participant type. The terminal-run cue is comparatively homogeneous for side choice ($\tau/|\hat{\beta}| = 1.07$ –1.34), so it should predict side choice at the population level. We test this prediction on the pilot sample ($n_{FN} = 9$, $n_{FP} = 5$),

which consists of different participants than the main sample (though drawn from the same participant pool). First, for betting: we compute the rule’s predicted betting rate from the pilot sequences – each imbalanced sequence counts as a “bet”, each perfectly balanced sequence as a coin flip (0.5) – and compare it to participants’ realized betting rate. The rule implies a predicted betting rate of 84.4%. In FN, realized betting of 87.5% (95% CI [82.1%, 92.2%], percentile bootstrap over participants) matches this prediction closely. In FP, realized betting of 95.6% ([88.1%, 100%]) exceeds it—unsurprisingly, since three of the five FP participants bet on every trial. Second, for side choice: we compare participants’ actual choices, trial by trial, against two rules – betting against the terminal run (on bettor trials where a run is present) and with the imbalance majority (on bettor trials with no run but a clear majority) – and test each rule’s accuracy against chance (0.5), since H and T are symmetric by design. Against the run, accuracy is 67.9% in FN (95% CI [60.1%, 76.2%], excluding chance) and 55.6% in FP ([37.0%, 74.7%], too imprecise at $n = 5$ to distinguish from chance). With the imbalance majority, accuracy is 54.1% in FN ([45.0%, 65.3%]) and 52.7% in FP ([40.4%, 62.4%])—both intervals contain chance, consistent with the weak and heterogeneous population effect of imbalance on side choice. Taken together, these out-of-sample results confirm both predictions: the imbalance cue forecasts whether participants bet but not which side they choose, while the terminal-run cue forecasts side choice – in FN; the FP pilot is too small to be informative. These are deliberately the simplest possible rules, with no information about individual participants; a model that first classifies participant type – from early-trial behavior, debrief strategy, or both – before predicting side choice would likely sharpen the imbalance prediction, and we leave developing one out of sample to future research.

The analyses presented in this section also answer the question with which it opened: whether the sequences that are economically costly are the same as those that steer directional choice. They are largely not. Across all 64 sequences, side bias and welfare loss are unrelated (posterior median $\rho = -0.06$, 95% CrI [-0.25, 0.14]); holding imbalance fixed, a 10-percentage-point stronger side bias is associated with at most 0.5pp more welfare loss. Table 8 shows why: the two outcomes respond to different features of the sequence. The terminal run shifts side choice but leaves betting, stakes, and welfare loss credibly unchanged. Imbalance draws participants into betting and weakens side bias; because betting is optimal in this treatment, moderate imbalance even reduces welfare loss, and only in the two pure sequences does the additional engagement come with inflated stakes and losses.²⁹

The dissociation extends to participants (Online Appendix Table C12). Participants whose side choices respond more strongly to the run and imbalance cues do not incur larger welfare losses; if anything, they incur smaller ones (confirmatory sample, $n = 95$: $\rho = -0.19$, 95% CrI [-0.40, 0.04], and $\rho = -0.22$, [-0.41, -0.00]), with the posterior probability of a negative association between 0.89 and 0.98 across both cues, samples, and treatments. The same holds for classes (Online Appendix Table D6): participants who credibly lean against imbalance lose 7.3% of the endowment, compared with 17.6% for those with no credible response to it (-10.0 pp, 95% CrI [-19.8, -0.4]; -6.3 pp, [-13.0, 1.8] relative to all others). The gap arises in staking rather than betting: their stakes deviate less from the optimum (-12.6 pp, [-22.1, -2.6]), while betting rates are similar (+0.8 pp, [-10.4, 12.4]). The preregistered models make this division explicit (Online Appendix Table C15): participants account for 98% of the between-participant and between-sequence variation in welfare loss and 94% in betting, but only 57% (95% CrI [43, 70]) in side choice, where sequences matter about as much as individuals. The gambler’s fallacy, usually treated as a costly error, is in this setting close to harmless: it pulls hardest on sequences whose imbalance is too mild to inflate betting and stakes, and is overridden precisely where imbalance is extreme enough to do economic damage. Put simply, the sequence largely determines *which* side

²⁹ In the FP treatment, side bias and welfare loss are likewise unrelated across sequences ($\rho = -0.09$, 95% CrI [-0.33, 0.16]); because betting is suboptimal at $m = 1.9$, welfare loss there rises monotonically with imbalance (Online Appendix Table C13).

Table 8: Sequence features and outcomes across the four margins

	Direction	Engagement and welfare		
	Side bias $ \hat{\mu}_s^h - \bar{h} $	Betting rate $\hat{\mu}_s^b$	Stake deviation $\hat{\mu}_s^a$	Welfare loss $\hat{\mu}_s^c$
Terminal run	5.6* [3.8, 7.6]	0.8 [-0.8, 2.3]	0.1 [-0.7, 0.9]	-0.4 [-1.0, 0.1]
Imbalance 2-4	-2.0* [-3.9, -0.1]	13.7* [11.9, 15.5]	-0.5 [-1.4, 0.5]	-1.7* [-2.4, -1.0]
Imbalance 1-5	-4.5* [-7.3, -1.8]	16.8* [14.4, 19.2]	3.1* [2.1, 4.4]	0.0 [-0.9, 0.9]
Imbalance 0-6	-9.2* [-13.3, -4.4]	21.4* [17.2, 25.2]	11.4* [9.0, 13.8]	15.5* [11.9, 19.9]
Constant (balanced, no run)	7.4* [5.6, 9.1]	63.6* [62.0, 65.2]	6.6* [5.7, 7.3]	13.0* [12.2, 14.2]
Sequences	64	64	64	64
Posterior draws	1000	1000	1000	1000

Notes: Each column regresses a sequence-level quantity on a terminal-run indicator (last two symbols identical) and imbalance-class indicators (reference: balanced 3-3), separately within each posterior draw. Cells report the posterior median of the coefficient (percentage points) with its 95% credible interval in brackets; an asterisk denotes an interval excluding zero. Side bias uses the full sample; betting rate, stake deviation, and welfare loss (shares of the endowment) use the confirmatory sample. The 0-6 class contains only the two pure sequences.

participants bet on, whereas the participant largely determines *how much* that bet costs.

4 Discussion

This article reported a laboratory experiment examining how a panel of all 64 unique six-toss coin histories affects behavior on the next toss (whether to bet, how much to stake, and which side to choose) and how much welfare it costs participants, measured against what an expected-utility maximizer with the same risk preference would have bet and staked. The result, compressed into one sentence, is: *The sequences that cost participants the most money are the ones they barely react to, and the sequences they react to the most cost them almost nothing.*

Unlike most discussions, we begin with what we did not get right: our preregistration expected the two pure sequences, HHHHHH and TTTTTT, to generate the clearest majority symbol, directionally consistent with gambler's fallacy. The data, however, show that side choice on both sequences is not credibly biased toward either side. These sequences, instead, are extreme in the confidence behind that choice, as measured by betting rates and amounts staked, resulting in the highest welfare loss in the panel, 30% and 26% of the endowment, respectively, compared to a sample mean of 12.5%. For reference, the closest runner-up, HHTTHT, generated a welfare loss of 16%. Importantly, the pure sequences differ structurally from the rest of the panel: their welfare loss is driven primarily by over-staking, while in the other sequences it stems predominantly from underbetting, with variation in betting being roughly twice as large as in staking.³⁰ For the betting rate, no sequence credibly cleared 100% in the FN treatment, where betting was EUT-optimal for all participants.³¹ For stake sizes, the average deviation from the optimal amount was +7.3% of the endowment, suggesting mild general over-staking. However,

³⁰ Sequence-level posterior means span 54.8% to 85.7% for betting rate (a spread of about 31 percentage points) against 3.5% to 19.8% of the endowment for stake deviation (about 16 percentage points).

³¹ Surprisingly, the betting rate difference of only 8.7pp from the FP treatment, where abstention from betting was EUT-optimal for all but two participants, is a pattern that EUT alone cannot explain, though boredom or multiplier-insensitivity are plausible candidates. The pilot's eye-tracking data at least rules out participants simply not seeing the payoff information, which was part of every decision screen.

the average conceals two roughly equal and opposite subpopulations, a split only partly attributable to risk preferences.³² This individual-level split is not an isolated feature of stake deviation: participant heterogeneity dominates sequence heterogeneity across the board, $\sigma_u/\sigma_s = 3.87$ for betting rate and 6.41 for welfare loss. With the two pure sequences the exception to this rule, *economic losses are mainly a matter of who participants are, not which sequence they face.*

Structurally, sequences can be characterized by the magnitude of their Heads-Tails imbalance and the presence of a terminal run of at least two symbols. These two features relate differently to betting rate and side bias magnitude. For the betting rate, moving from balanced to fully imbalanced sequences raises it by 21.1pp (Cohen’s $h = 0.36$), whereas a run does not credibly affect it. For side bias magnitude, the same move reduces it by 9.2pp, whereas a run raises it by 5.6pp. Regarding the direction of side bias, the run elicits robust gamblerishness, that is, bets on the side opposite to the run: a terminal run lowers the probability of choosing its symbol by 13.6pp ($|h| = 0.27$), and 30 of the 32 run-ending sequences elicit bets against it. The rule also holds out of sample, predicting 67.9% (95% CI [60.1, 76.2]) of bets on such sequences in the pilot’s FN treatment, beating the 50% chance level. Imbalance weakly elicits hot-handishness, as more of the same symbol prompts the same side choice: moving from a balanced to a pure sequence raises the probability of choosing the majority symbol by 8.0pp ($h = 0.16$). The two pulls differ in how uniformly participants share them. The run pulls in one direction only: 40% of participants bet against the terminal run, 1% with it, and 59% are uncertain. Imbalance splits participants: as the imbalance grows, 19% choose the minority symbol (balance-restorers), 30% the majority symbol (majority-followers), and 51% are uncertain.³³ The majority-followers are of particular interest: 33% of them credibly bet against the run, against 61% of balance-restorers, and they do so only about half as strongly (median slope -0.74 vs. -1.53). This largely reflects their reaction to imbalance: the share of their bets against the terminal run falls from 68% on balanced sequences to 32% at the 4–2 split and to 9% in the pure sequences.³⁴

To interpret these opposing forces, we turn to the strand of literature showing that reversal dominates after short streaks and continuation after long ones (Asparouhova et al., 2009; Rabin, 2002; Rabin & Vayanos, 2010). As our sequences have six symbols, we cannot test this directly, but the two cues map onto it, the terminal run being a short streak and imbalance a property of the whole sequence, and we offer a theoretical explanation for the run: to conserve effort, participants rely on the terminal run instead of evaluating the whole structure, a cheap cue that is informative because run-ending sequences are, as a rule, more likely to be imbalanced (see footnote 26). Such reliance is plausible given the pace of decisions: participants take a median of 5.6 seconds (mean 7.2) to read the sequence, choose a side, and set a stake, which leaves little time to evaluate its whole structure. This does not make the run rule a snap reflex: participants who take longer to decide bet against the run more strongly ($\rho = -0.29$, 95% CrI $[-0.38, -0.18]$), which fits a simple but deliberate rule rather than a quick associative response (Sloman, 1996), although the association is correlational and faster decisions may also reflect participants who engage less with the task. This is a form of *satisficing* (Simon, 1956): among participants with a discernible strategy, balance restoration is the default, and because the terminal run usually also consists of the majority symbol, betting against it restores balance both locally, within the short terminal streak,

³² Individual risk parameters explain roughly a third of the variance in stake deviation, leaving most of it unaccounted for. And it is the confident group, not the cautious one, that pays for it: the 43% of participants classified as solid welfare losers are almost exclusively frequent bettors whose losses trace back to miscalibrated stake size rather than to abstaining from betting.

³³ Debrief reports corroborate this classification. Among the 97 participants who described a rule, 14 of the 17 who reported betting against the trend counter imbalance, 3 of the 4 who reported betting with it follow imbalance, and all 12 who reported no trend-based strategy are classified as uncertain.

³⁴ Classifying participants on sequences without a terminal run (binomial test of majority-symbol choice, $p < .05$), so that classification and outcome rest on different trials, gives the same pattern: majority-followers bet against the run on 72% of balanced run-ending sequences, but choose its side on 81% of the pure ones. These classes are based on raw choices and differ from the model-based classes above.

and globally, across the whole sequence.³⁵ There is, however, the group of majority-followers on the imbalance cue (30% of participants), who rely on the run cue when the sequence is balanced but abandon it as imbalance grows. This pattern is consistent with these participants gradually coming to disbelieve that the generator is random and switching to majority-following, in line with the literature on the perceived generator (Ayton & Fischer, 2004; Burns & Corpus, 2004; Caruso et al., 2010): generators perceived as random elicit reversal, whereas nonrandom or intentional ones induce continuation. Such a switch is plausible because strongly imbalanced sequences are also easier to read at first glance: the excess of one symbol is salient and makes the sequence look nonrandom. In our data, imbalance-followers report weaker belief in independence than participants with no discernible response to imbalance (mean difference -2.8 points, 95% CI $[-3.8, -1.8]$, treatments pooled), whereas this belief is similar across run classes (against-run vs. uncertain: -0.2 $[-1.1, 0.8]$). This account yields a testable prediction for future research: if correct, reducing the salience of imbalance, for instance by presenting histories sequentially, showing only the most recent tosses, or limiting exposure time, should lead fewer participants to follow the majority and more to bet against the run. Whether the same holds in longer sequences is less clear: imbalance is harder to take in at a glance, which favors reliance on the run, but a terminal run becomes a less informative cue to imbalance, which works against it.

So far, we have discussed the proximate reasons for reversal and continuation tendencies in detail; here we turn briefly to the ultimate explanation, that is, how these fallacies could contribute to survival and thus why they persist in human cognition. Although we find more support for gamblerish than hot-handish tendencies, in line with Oskarsson et al. (2009), and a within-person correlation between the two (run- and imbalance-slopes: Spearman $\rho = 0.22$, $p = .007$, FN), in line with Sundali and Croson (2006), the room for sequence-related effects on welfare is small: individual heterogeneity accounts for 98% of the variation in welfare loss, whereas sequence effects account for little. Side choice, however, does respond to the sequence: its variation is divided roughly equally between participants (57%) and sequences (43%). The natural question is what advantage, if any, these rather complicated side-choice rules offer over more effortless ones, such as picking at random, beyond the evolutionary accounts of pattern-seeking.³⁶ The data hint that the advantage is engagement. Participants with no discernible response to either cue lose more of their endowment than all other participants: 9.4pp more on average (95% CrI $[1.2, 23.2]$), but only 2.8pp more at the median ($[-0.4, 8.4]$), which points to a minority of heavy losers. The gap arises in staking rather than betting (stake deviation $+10.5$ pp, $[2.0, 21.2]$; betting rate $+1.8$ pp, $[-11.2, 12.2]$), possibly because they place somewhat more of their stakes at focal values of 50 and 100 ECU ($+13.5$ pp, $[-0.9, 29.6]$; not replicated in FP). They also decide 1.6 s (26%) faster ($[-2.4, -0.8]$; replicated in FP). It looks as if participants who try to extract information from the random sequence stake more cautiously, although the mechanism behind this is unclear and warrants future research.

Our study has several limitations. Most importantly, behavior differed surprisingly little between the two treatments that vary the multiplier on staked amounts: betting rates and stakes were similar in FP, where EUT prescribes abstention for all but two participants, and in FN, where it prescribes betting for all. We cannot explain this gap between prediction and behavior. Whether it reflects insensitivity to the multiplier or an urge to act regardless of the odds (almost half of participants report betting often

³⁵ In our panel of 64 sequences, the run is the majority symbol in 22 of the 24 imbalanced sequences with a terminal run (14 of 14 for runs of three or more). In the two exceptions, HHHHTT and TTTTHH, the run is the minority symbol, yet participants bet against it about as often as elsewhere (55% of 316 bets, versus 56.5% across all sequences). This is consistent with the terminal run cue being applied irrespective of whether it restores balance globally, as one expects of a heuristic (Tversky & Kahneman, 1974): the rule is mostly useful, but sometimes backfires.

³⁶ Foster and Kokko (2009) show that selection favors assigning causality between events as long as occasional correct detections carry large fitness benefits, even when most detections are wrong, and that this can extend to events that are in fact unrelated when organisms are limited in computing the underlying probabilities. Error management theory (Johnson et al., 2013) adds an economic layer: when a false negative is costlier than a false positive, selection favors biases that err toward false positives. Seeing a pattern where there is none is cheap, whereas missing a real one is not: fleeing from a bush rustled by the wind costs far less than staying put and discovering it was a lion.

mainly because doing something felt better than not betting, and this item is strongly related to betting rates in both treatments), the methodological lesson is that, at these stakes, the prospect of winning did little to move choices, and our findings may not carry over to settings in which more is at stake. In addition, FP has only 53 participants, so statements about replication in that treatment are imprecise. Second, participants viewed each six-toss history as a static, completed sequence rather than experiencing it unfold toss by toss. For beliefs this is less restrictive than it may seem, since streaks are perceived in presented sequences of outcomes as well (Gilovich et al., 1985), but it leaves open whether betting and staking respond in the same way when histories unfold in real time and/or as actual realizations. Moreover, the strength of the gambler’s fallacy in betting depends on how many past outcomes are displayed (Wang & Li, 2025), so with histories fixed at six tosses we cannot say whether our patterns hold for shorter or longer ones, nor locate the streak length at which behavior switches. Both points relate to the salience of the sequence discussed above: with the full history displayed at once, imbalance is as salient as it can be, and sequential presentation or a shorter display are the natural tests of our prediction that lowering its salience makes side choice more gamblerish. Third, we infer the fallacies from side choices alone and did not elicit beliefs, neither of those who bet nor of those who refrained from betting, whose side choice we never observe. Because the coin is fair and the side carries no payoff consequence, a bet against a run need not reflect a belief that reversal is likely, so our interpretations in terms of beliefs about the generator remain inferences. Beyond the avenues for future research suggested above, our findings should be replicated on a more diverse sample with a more balanced sex ratio, and on actually realized tosses rather than pre-made histories.

To close where we began, we make no claim that our design replicates the Monte Carlo wheel. Our streaks last six tosses, not twenty-six; they are presented as histories rather than observed sequentially; the stakes are small; and only the unfavorable treatment, whose multiplier happens to match an outside bet on an American roulette wheel, carries anything like a house edge. Nor can we speak to the crowd: our participants decide individually, so whatever herding or bandwagon effects may have shaped the mood at the table in 1913 are absent from our design. With these caveats, the legend acquires a modest moral. Faced with an extreme streak, our participants did not systematically bet on the overdue side, but they did bet more and stake more. Half of them would have won and half lost regardless of the side chosen, yet the house, which needs no fallacy to collect its edge, gains from every additional franc wagered.

Author Statement

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Conflict of Interest

The authors declare no conflict of interest.

Supplementary Material

An Online Appendix for this article can be found at ...

Data Availability

The data and code underlying this article are available in ...

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